

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Introduction of Coordination Compounds

- Which of the following statements is incorrect ?
 (A) Coordination compounds and complexes are synonymous terms.
 (B) Complexes must give ions in the solution.
 (C) Complexes may give ions in the solution or may not give ions in the solution.
 (D) Generally complex ion does not dissociate into its component parts even in the solution.
 CCC093
- Which will not give test of all the ions present in it
 (A) $K_2Fe_2(SO_4)_4 \cdot 24H_2O$ (B) $(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O$
 (C) $K_3[Fe(CN)_6]$ (D) $Fe_2(SO_4)_3$
 CCC094
- Statement-I : $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$ is a double salt compound.
 Because
 Statement-II : It ionises to give a complex ion.
 These questions contains, Statement-I (assertion) and Statement-II (reason).
 (A) Statement-I is True, Statement-II is True ; Statement-II is a correct explanation for Statement-I
 (B) Statement-I is True, Statement-II is True ; Statement-II is NOT a correct explanation for Statement-I
 (C) Statement-I is True, Statement-II is False.
 (D) Statement-I is False, Statement-II is True.
 CCC100
- The number of ions formed on dissolving one molecule of $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ in water is :
 (A) 4 (B) 5 (C) 3 (D) 6
 CCC107

Terminology Used in Coordination Compounds

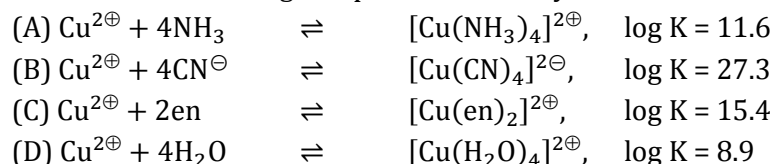
- The coordination number of Cr in the complex ion $[Cr(H_2O)_6]^{3+}$ is :-
 (A) Zero (B) 3 (C) 6 (D) None of these
 CCC096
- All ligands are :-
 (A) Lewis acid (B) Lewis base (C) Neutral (D) None of these
 CCC097
- The co-ordination number of a central metal atom in a complex is determined by
 (A) the number of ligands around a metal ion bonded by sigma and pi-bonds both
 (B) the number of ligands around a metal ion bonded by pi-bonds
 (C) the number of ligands around a metal ion bonded by sigma bonds
 (D) the number of only anionic ligands bonded to the metal ion.
 CCC105
- Ligand in a complex salt are
 (A) anions linked by coordinate bonds to a central metal atom or ion
 (B) cations linked by coordinate bonds to a central metal or ion
 (C) molecules linked by coordinate bonds to a central metal or ion
 (D) ions or molecules linked by coordinate bonds to a central atom or ion
 CCC106

9. The total number of monodentate ligands attached to the central metal ion through coordinate (sigma) bond is called.
 (A) Valency of the metal ion (B) Oxidation state of the metal ion
 (C) Coordination number of metal ion (D) None of the above

CCC115

Formation Constant and Dissociation Constant

10. Which of the following complexes formed by Cu^{2+} ions is most stable ?



CCC095

11. For the reaction of the type $\text{M} + 4\text{L} \rightleftharpoons \text{ML}_4$
 (A) larger the stability constant, lower the proportion of ML_4 that exists in solution.
 (B) larger the stability constant, higher the proportion of ML_4 that exists in solution.
 (C) smaller the stability constant, higher the proportion of ML_4 that exists in solution.
 (D) None of the above

CCC101

12. Consider the following reactions.



On the basis of reactions above which of the following is correct ?

- (A) Overall stability constant = $K_1 K_2 K_3$
 (B) $K_1 K_2 K_3 = \frac{[\text{XY}_3]}{[\text{X}][\text{Y}]^3}$
 (C) Dissociation constant = $\frac{1}{\text{Formation constant}}$.
 (D) All of the above are correct.

CCC103

Types of Ligands: On the basis of Charge

13. In the coordination compound $\text{K}_4[\text{Ni}(\text{CN})_4]$, the oxidation state of nickel is –
 (A) 0 (B) +1 (C) +2 (D) –1

CCC099

Types of Ligands: On the basis of Denticity; Ligand as Polydentate Ligand; Chelating Ligands; Ambidentate Ligands; Flexidentate Ligands;

14. The coordination number of cobalt in $[\text{Co}(\text{en})_2\text{Br}_2]\text{Cl}_2$ is :-
 (A) 2 (B) 4 (C) 6 (D) 8

CCC098

15. Bidentate ligand is :-
 (A) $\text{CN}^\ominus$ (B) Ethylene diamine (C) $\text{EDTA}^{4\ominus}$ (D) $\text{SCN}^\ominus$ CCC113
16. Triphenyl phosphine is :-
 (A) Neutral and monodentate ligand (B) Neutral and tridentate ligand
 (C) Uninegative and unidentate ligand (D) Trinegative and tridentate ligand CCC114
17. Which of the following ligand does not act as bidentate ligand
 (A) dipy (B) dien (C) $\text{gly}^\ominus$ (D) $\text{dmg}^\ominus$ CCC116
18. Diethylene triamine is:
 (A) Chelating agent (B) Polydentate ligand
 (C) Tridentate ligand (D) All of these CCC117
19. $\text{C}^\ominus\text{N}$ is an ambidentate ligand because
 (A) it has multiple bonds (B) carbon as a negative charge
 (C) it form chelate (D) both C & N can donate lone pair CCC118
20. Which of the following ligands may be ambidentate ligand –
 (A) $\text{NO}_2^\ominus$ (B) $\text{SCN}^\ominus$ (C) Both (A) and (B) (D) None of these CCC119
21. An ambident ligand is one which
 (A) is linked to the metal atom through two donor atom
 (B) has two donor atoms, and only one of them has the capacity to form a coordinate bond [or a sigma (σ) bond]
 (C) has two donor atoms, but either of the two can form a coordinate bond
 (D) forms chelate rings. CCC122
22. Which of the following does not form a chelate ?
 (A) EDTA (B) Oxalate (C) Pyridine (D) Ethylenediamine CCC124
23. Identify the correct statements for the behaviour of ethane-1, 2-diamine as a ligand.
 (i) It is a neutral ligand. (ii) It is a bidentate ligand.
 (iii) It is a chelating ligand. (iv) It is a unidentate ligand.
 (A) (i), (ii) and (iii) (B) (ii) and (iii)
 (C) (iii) and (iv) (D) (i), (iii) and (iv) CCC125
24. The coordination number and the oxidation state of the element 'E' in the complex $[\text{E}(\text{en})_2(\text{C}_2\text{O}_4)]\text{NO}_2$ (where (en) is ethylene diamine) are, respectively,
 (A) 6 and 2 (B) 4 and 2 (C) 4 and 3 (D) 6 and 3 CCC104

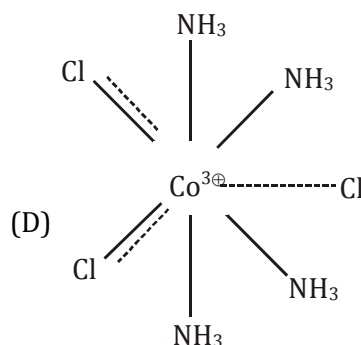
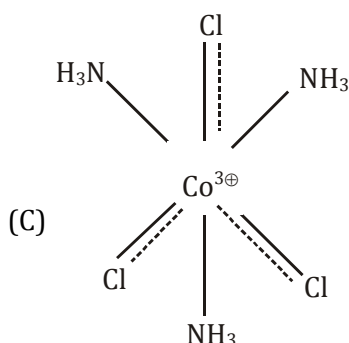
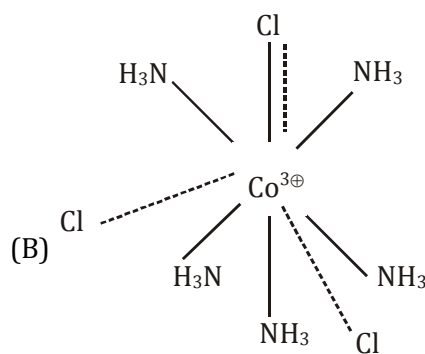
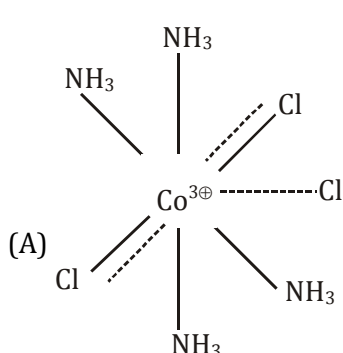
Organic Ligands

25. How many π electrons are donated by $\text{C}_5\text{H}_5^\ominus$ ligand -
 (A) 2 (B) 4 (C) 5 (D) 6

CCC161

Werner Theory; Electrical and Molar Conductivity

26. Consider the following statements:
 According to the Werner's theory.
 (a) Ligands are connected to the metal ions by covalent bonds.
 (b) Secondary valencies have directional properties
 (c) Secondary valencies are non-ionisable
 (A) a, b and c are correct (B) b and c are correct
 (C) a and c are correct (D) a and b are correct
27. Which of the following complex will give white precipitate with barium chloride solution :-
 (A) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ (B) $[\text{Cr}(\text{NH}_3)\text{SO}_4]\text{Cl}$
 (C) $[\text{Co}(\text{NH}_3)_6]\text{Br}_3$ (D) None of these
28. A complex of platinum, ammonia and chloride produces four ions per molecule in the solution. The structure consistent with the observation is :
 (A) $[\text{Pt}(\text{NH}_3)_4]\text{Cl}_4$ (B) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$
 (C) $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$ (D) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$
29. Which of the following Werner's complex has least electrical conductivity ?



CCC144

Coordination Compounds

30. Which of the following has least conductivity in aqueous solution.
 (A) $\text{CoCl}_3 \cdot 4\text{NH}_3$ (B) $\text{CoCl}_3 \cdot 3\text{NH}_3$ (C) $\text{CoCl}_3 \cdot 5\text{NH}_3$ (D) $\text{CoCl}_3 \cdot 6\text{NH}_3$ CCC145
31. Which gives only 50% mole of AgCl , when reacts with AgNO_3 :-
 (A) $\text{PtCl}_2 \cdot 4\text{NH}_3$ (B) $\text{PtCl}_4 \cdot 5\text{NH}_3$ (C) $\text{PtCl}_4 \cdot 4\text{NH}_3$ (D) $\text{PtCl}_4 \cdot 3\text{NH}_3$ CCC146
32. Number of electrolytes in the complex $\text{CoCl}_3 \cdot 5\text{NH}_3$
 (A) 3 (B) 2 (C) 4 (D) 5 CCC147
33. A co-ordination complex of cobalt has molecular formula containing five ammonia molecules, one nitro group and two chlorine atoms for one cobalt atom. One mole of this compound produces three mole ions in an aqueous solution. In reacting this solution with excess of silver nitrate solution, two moles of AgCl get precipitated. The ionic formula of this complex would be :-
 (A) $[\text{Co}(\text{NH}_3)_4 \cdot \text{NO}_2 \text{Cl}] \cdot [(\text{NH}_3)\text{Cl}]$ (B) $[\text{Co}(\text{NH}_3)_5 \text{Cl}] \cdot [\text{Cl}(\text{NO}_2)]$
 (C) $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}_2$ (D) $[\text{Co}(\text{NH}_3)_5] \cdot [(\text{NO}_2)_2 \text{Cl}_2]$ CCC148
34. What is the secondary valency of following compounds $\text{PtCl}_2 \cdot 2\text{NH}_3$, $\text{CoCl}_3 \cdot 4\text{NH}_3$ and $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ if moles of AgCl precipitated per mole of the given compounds with excess AgNO_3 respectively are: 0, 1 and 2
 (A) 6, 4, 4 (B) 4, 6, 6 (C) 4, 4, 6 (D) 2, 4, 6 CCC149
35. Correct formula of the compound which gives a white precipitate with BaCl_2 solution, but not with AgNO_3 solution is.
 (A) $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ (B) $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ (C) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Br}_2$ (D) $[\text{Pt}(\text{NH}_3)_4\text{Br}_2]\text{Cl}_2$ CCC150
- Effective Atomic Number; Calculation of EAN; Sidwick EAN Rule; Effective Atomic Number of Bridging Complexes; Oxidising and Reducing Property of Carbonyl Complex**
36. The EAN of cobalt in the complex ion $[\text{Co}(\text{en})_2\text{Cl}_2]^\oplus$ is :-
 (A) 27 (B) 36 (C) 33 (D) 35 CCC154
37. If EAN of a central metal ion $\text{X}^{2\oplus}$ in a complex is 34. and atomic number of X is 28. The number of monodentate ligands present in complex are:-
 (A) 3 (B) 4 (C) 6 (D) 2 CCC155
38. Which of the following does not obey Sidgwick effective atomic number rule -
 (A) $\text{Co}_2(\text{CO})_8$ (B) $[\text{Fe}(\text{CO})_5]$ (C) $[\text{V}(\text{CO})_6]^\ominus$ (D) $[\text{Mn}(\text{CO})_6]$ CCC156
39. The complex that violates the EAN rule :-
 (A) Potassium ferrocyanide (B) Potassium ferricyanide
 (C) Tetra carbonyl Nickel (D) Cobalt (III) hexamine chloride CCC157
40. The EAN of metal atoms in $\text{Fe}(\text{CO})_2(\text{NO})_2$ and $\text{Co}_2(\text{CO})_8$ respectively are
 (A) 34, 35 (B) 34, 36 (C) 36, 36 (D) 36, 35 CCC158

41. Effective atomic number of $\text{Co}(\text{CO})_4$ is 35, hence it is less stable. It attains stability by
(A) Oxidation of Co (B) Reduction of Co (C) Dimerization (D) Both (B) & (C) CCC159
42. Among the following complexes which can act as oxidising agent.
(A) $[\text{Mn}(\text{CO})_6]$ (B) $[\text{Mn}(\text{CO})_6]^\oplus$ (C) $[\text{Mn}(\text{CO})_5]^\ominus$ (D) $[\text{V}(\text{CO})_6]$ CCC160
43. Ligands are neutral or ionic species capable of donating at least one electron pair to central metal. Hence ligands can be of different denticities.
 $[\text{Mn}(\text{CO})_5]$ can attain more stability by :
(A) Oxidation of itself (B) Reduction of itself
(C) Dimerization (D) Both (B) and (C) CCC165
44. Ligands are neutral or ionic species capable of donating at least one electron pair to central metal. Hence ligands can be of different denticities.
Which of the following complexes follow Sidgwick EAN rule ?
(A) $[\text{Fe}(\eta^5 - \text{C}_5\text{H}_5)_2]$ (B) $[\text{K}[\text{PtCl}_3(\eta^2 - \text{C}_2\text{H}_4)]]$
(C) $[\text{V}(\text{CO})_6]$ (D) $[\text{Mn}(\text{CO})_6]$ CCC166

IUPAC Nomenclature of Coordination Compounds

45. $\text{K}_3[\text{Fe}(\text{CN})_6]$ is :-
(a) Potassium hexacyanoferrate(II) (b) Potassium hexacyanoferrate(III)
(c) Potassium ferricyanide (d) Hexacyanoferrate(III) potassium
Correct answer is :-
(A) Only (a) and (b) (B) Only (b) and (c) (C) Only (a) and (c) (D) Only (b) and (d) CCC172
46. The IUPAC name of $\text{Fe}(\text{CO})_5$ is
(A) Pentacarbonyl ferrate(0) (B) Pentacarbonyl ferrate(III)
(C) Pentacarbonyliron(0) (D) Pentacarbonyliron(II) CCC173
47. IUPAC name of $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ is
(A) Tetraamminedichloridochromium(III) chloride
(B) dichloridotetraamminechromium(III) chloride
(C) trichloridotetraamminechromium(III)
(D) Chromiumtetraamminetrichloride CCC174
48. The IUPAC name for the coordination compound $\text{Ba}[\text{BrF}_4]_2$ is
(A) Barium tetrafluorobromate (V) (B) Barium tetrafluorobromate (III)
(C) Barium bis (tetrafluorobromate) (III) (D) none of these CCC175
49. Give the IUPAC name of the complex compound $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}](\text{NO}_3)_2$.
(A) BromoaquatetraamineCobalt(III) nitrate
(B) BromoaquatetraaminoCobalt(III) nitrate
(C) BromoaquatetraammineCobaltate(III) nitrate
(D) TetraammineaquabromoCobalt(III) nitrate CCC176

50. The formula for the compound tris (ethane-1, 2-diamine)cobalt(III) sulphate is
 (A) $[\text{Co}(\text{en})_3]\text{SO}_4$ (B) $[\text{Co}(\text{SO})_4(\text{en})_3]$
 (C) $[\text{Co}(\text{en})_3](\text{SO}_4)_2$ (D) $[\text{Co}(\text{en})_3]_2(\text{SO}_4)_3$ CCC178
51. The number of ions formed, when bis(ethane-1,2-diamine)copper(II) sulphate is dissolved in water will be
 (A) 1 (B) 2 (C) 3 (D) 4 CCC179
52. The IUPAC name for $[\text{Co}(\text{NCS})(\text{NH}_3)_5]\text{Cl}_2$ is :-
 (A) Pentaammine(thiocyanato-N)cobalt(III) chloride
 (B) Pentaammine(thiocyanato-S)cobalt(III) chloride
 (C) Pentaammine(isothiocyanato-N,S)cobalt(III) chloride
 (D) Pentaammine(mercapto-N)cobalt(III) chloride CCC181
53. The IUPAC name for $[\text{Co}(\text{NH}_3)_6][\text{Cr}(\text{CN})_6]$ is :-
 (A) Hexaamminecobalt(III) hexacyanochromate(III)
 (B) Hexacyanochromium cobalt hexaammine(VI)
 (C) Hexaammine cobalt(III) hexacyanochromium(VI)
 (D) Hexacyanochromium(III) hexaammine cobalt(III) CCC182
54. Name the following compound according to IUPAC system

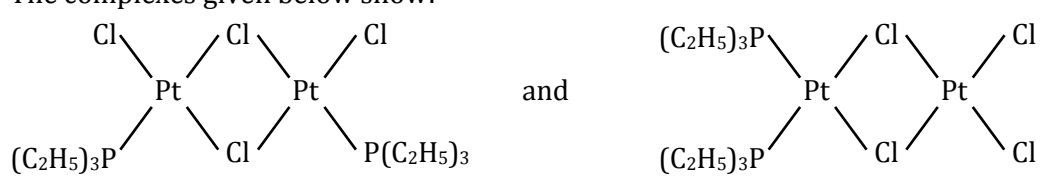
$$\left[\begin{array}{c} \text{Cl} \\ \diagup \quad \diagdown \\ (\text{NH}_3)_4 \text{Co} \quad \text{Co}(\text{NH}_3)_4 \\ \diagdown \quad \diagup \\ \text{NO}_2 \end{array} \right] (\text{SO}_4)_2$$

 (A) Octammine- μ - chloro - μ -nitrodicobalt(II) sulphate
 (B) Octaammine- μ - chloro - μ -nitrodicobalt(III) sulphate
 (C) Octaammine- μ - nitro - μ -chlorodicobalt(III) sulphate
 (D) None of the above CCC186

IUPAC Nomenclature of Some Specific Complexes and Common Name

55. The IUPAC name of the Wilkinson's catalyst $[\text{RhCl}(\text{PPh}_3)_3]$ is
 (A) Chloridotris(triphenylphosphine)rhodium(I)
 (B) Chloridotris(triphenylphosphine)rhodium(IV)
 (C) Chloridotris(triphenylphosphine)rhodium(0)
 (D) Chloridotris(triphenylphosphine)rhodium(VI) CCC180
56. The compound $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}^\oplus]$ is called
 (A) Sodium pentocyanonitrosoniumferrate(II)
 (B) Sodium nitroprusside
 (C) Sodium nitrosoferrocyanide
 (D) A & B both CCC183
57. The correct IUPAC name of the complex $[\text{Fe}(\pi\text{-C}_5\text{H}_5)_2]$ is
 (A) Cyclopentadienyliron(II) (B) Bis(η^5 -cyclopentadienyl)iron(II)
 (C) Dicyclopentadienylferrate(II) (D) Ferrocene CCC184

Structural Isomerism

58. Which of the following complex shows ionization isomerism.
 (A) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (B) $[\text{Cr}(\text{en})_2]\text{Cl}_2$ (C) $[\text{Cr}(\text{en})_3]\text{Cl}_3$ (D) $[\text{CoBr}(\text{NH}_3)_5]\text{SO}_4$
 CCC193
59. The compound $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ and $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ represent
 (A) Linkage isomerism (B) Hydrate isomerism
 (C) Ligand isomerism (D) Ionisation isomerism
 CCC195
60. Find the name of the hydrate isomer of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$, which is having lowest electrical conductivity excluding zero value of conductivity.
 (A) Hexaaquachromium(III)chloride
 (B) Tetraaquadichloridochromium(III)chloride dihydrate
 (C) Pentaquachloridochromium(III)chloride monohydrate
 (D) Triaquatrchloridochromium(III)chloride trihydrate
 CCC196
61. Which of the following compound cannot show coordination isomerism?
 (A) $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$ (B) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2][\text{Cr}(\text{NH}_3)_2(\text{C}_2\text{O}_4)_2]$
 (C) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2][\text{Pt}(\text{SCN})_2\text{Cl}_2]$ (D) $[\text{Pt}(\text{en})_2]^{2+}$
 CCC197
62. The type of isomerism present in pentaamminenitrochromium(III) chloride is :
 (A) Coordination (B) linkage
 (C) hydrate (D) Coordination position
 CCC198
63. The two compounds sulphatopenta-amminecobalt(III) bromide and penta-amminebromo cobalt(III) sulphate represent :
 (A) Linkage isomerism (B) Ionisation isomerism
 (C) Co-ordination isomerism (D) Hydrate isomerism
 CCC199
64. Which of the following complex may show atleast three type of isomerism, namely ionisation, hydrate and linkage :
 (A) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{SO}_4$ (B) $[\text{Ag}(\text{NH}_3)_2][\text{Ag}(\text{CN})_2]$
 (C) $[\text{Cr}(\text{H}_2\text{O})_5(\text{NO}_2)]\text{SO}_4$ (D) $[\text{Ni}(\text{H}_2\text{O})_6]\text{SO}_4$
 CCC200
65. The complexes given below show:

 (A) Optical isomerism (B) Co-ordination isomerism
 (C) Geometrical isomerism (D) Co-ordination position isomerism
 CCC201
66. The type of isomerism present in pentaamminenitrochromium (III) chloride is :
 (A) optical (B) linkage (C) hydrate (D) polymerisation
 CCC221

Stereoisomerism for Square Planar Complexes

67. The complex, $[\text{Pt}(\text{py})(\text{NH}_3)\text{BrCl}]$ will have how many geometrical isomers ?
 (A) 3 (B) 4 (C) 0 (D) 2 CCC188
68. How many coordination isomers of $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$ show geometrical isomerism.
 (A) All (B) One (C) Two (D) None CCC202
69. How many stereoisomers are possible for the square-planar complex $[\text{PtCl}_3\text{Br}]^{2-}$
 (A) 4 (B) 5 (C) 8 (D) None of these CCC335

Stereoisomerism for Tetrahedral Complexes

70. The tetrahedral complex $[\text{M}(\text{A})(\text{B})(\text{X})(\text{Y})]$, where A,B,X and Y are different ligands and M is a metal ion is
 (A) optically inactive (B) rotate plane polarized light
 (C) incomplete information (D) can't be said CCC187
71. Tetrahedral complexes generally do not exhibit geometrical isomerism because :
 (A) They have plane of symmetry.
 (B) They have centre of symmetry.
 (C) All positions about central atom are adjacent with respect to each other.
 (D) All position about central atom are at 180° with respect to central atom. CCC205
72. Which of the following tetrahedral complex has two stereoisomers ?
 [M = central atom, a, b, c, d = monodentate ligands]
 (A) $[\text{Ma}_2\text{b}_2]$ (B) $[\text{Ma}_2\text{bc}]$ (C) $[\text{Mab}_2\text{c}]$ (D) $[\text{Mabcd}]$ CCC206
73. A co-ordination compound $[\text{Mabcd}]$ has total two stereoisomers. The expected geometry of this complex is : [M = central atom, a, b, c, d = monodentate ligands]
 (A) Square planar (B) octahedral (C) Tetrahedral (D) trigonal bipyramidal CCC207

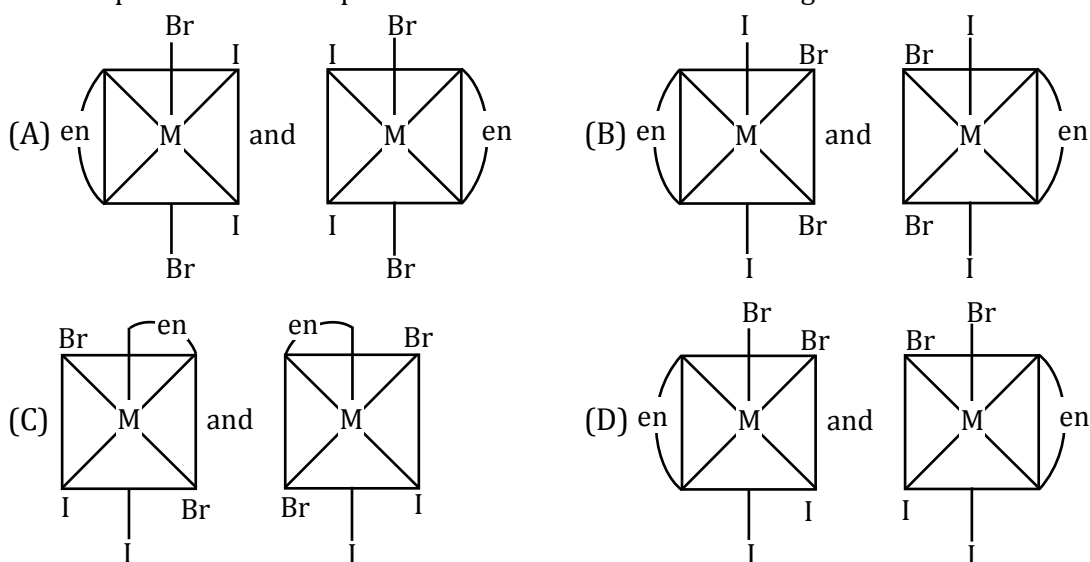
Stereoisomerism for Octahedral Complexes for Monodentate Ligand

74. Among the following coordination compounds/ ions
 (i) $[\text{Fe}(\text{CN})_6]^{3-}$ (ii) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (iii) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (iv) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
 Which species exhibit geometrical isomerism ?
 (A) (ii) only (B) (i) and (ii) (C) (ii) and (iv) (D) (i) and (iii) CCC210
75. Theoretically the Number of geometrical isomers expected for octahedral complex $[\text{Mabcdef}]$ is:-
 (A) Zero (B) 30 (C) 15 (D) 9 CCC215
76. Which of the following complex produce precipitates with AgNO_3 and exist in two geometrical isomeric form
 (A) $\text{PtCl}_2 \cdot 4\text{NH}_3$ (B) $\text{PtCl}_2 \cdot 3\text{NH}_3$ (C) $\text{PtCl}_4 \cdot 4\text{NH}_3$ (D) $\text{PtCl}_4 \cdot 2\text{NH}_3$ CCC216

77. The pair of molecules that exhibit geometrical isomerism are
 (A) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^{\oplus}$ and $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
 (B) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]^{2\oplus}$ and $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2\oplus}$
 (C) $[\text{Cr}(\text{NH}_3)_6]^{3\oplus}$ and $[\text{Co}(\text{NH}_3)_6]^{3\oplus}$
 (D) All of the above show geometrical isomerism CCC217
78. Which of the following complex shows optical isomerism -
 (A) $[\text{Cd}(\text{CN})_4]^{2\ominus}$ (B) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
 (C) $[\text{Zn}(\text{gly})_2]$ (D) $[\text{Ni}(\text{dmg})_2]$ CCC218
79. Which of the following can exhibit geometrical isomerism ?
 (A) $[\text{MnBr}_4]^{2\ominus}$ (B) $[\text{Pt}(\text{NH}_3)_3\text{Cl}]^{\oplus}$ (C) $[\text{PtCl}_2(\text{P}(\text{C}_2\text{H}_5)_3)_2]$ (D) $[\text{Fe}(\text{H}_2\text{O})_5\text{NOS}]^{2\oplus}$ CCC219

Stereoisomerism for Octahedral Complexes for Symmetrical Bidentate Ligand

80. Which of the following statements is/are true
 (A) In $[\text{PtCl}_2(\text{NH}_3)_4]^{2\oplus}$ complex ion, the cis-form is optically active, while trans-form is optically inactive
 (B) In $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3\ominus}$, geometrical isomerism does not exist, while optical isomerism exists
 (C) $[\text{Mabcd}]^{\pm}$ square planar complexes exhibit both optical as well as geometrical isomerism
 (D) In $[\text{Mabcd}]^{n\pm}$ tetrahedral complexes, optical isomerism cannot be observed CCC203
81. The number of optical isomers possible for $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3\ominus}$ is :
 (A) Three (B) Two (C) One (D) Four CCC209
82. Find complex which have maximum number of stereoisomers -
 (A) $[\text{Ma}_3\text{b}_3]$ (B) $[\text{Ma}_3\text{b}_2\text{c}]$ (C) $[\text{Ma}_2\text{b}_2\text{c}_2]$ (D) $[\text{M}(\text{AA})_2\text{b}_2]$ CCC222
83. The complex ion has two optical isomers. Their **CORRECT** configurations are:



84. Which of the following is not optically active :-
 (A) $[\text{Co}(\text{en})_3]^{3\oplus}$ (B) $[\text{Cr}(\text{ox})_3]^{3\ominus}$
 (C) cis - $[\text{CoCl}_2(\text{en})_2]^{\oplus}$ (D) trans - $[\text{CoCl}_2(\text{en})_2]^{\oplus}$ CCC226

CCC224

85. **Statement-1:** Complexes containing three bidentate groups such as $[\text{Cr}(\text{ox})_3]^{3-}$ and $[\text{Co}(\text{en})_3]^{3+}$ do not show optical activity.

Statement-2: Octahedral complex, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ shows geometrical isomerism.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

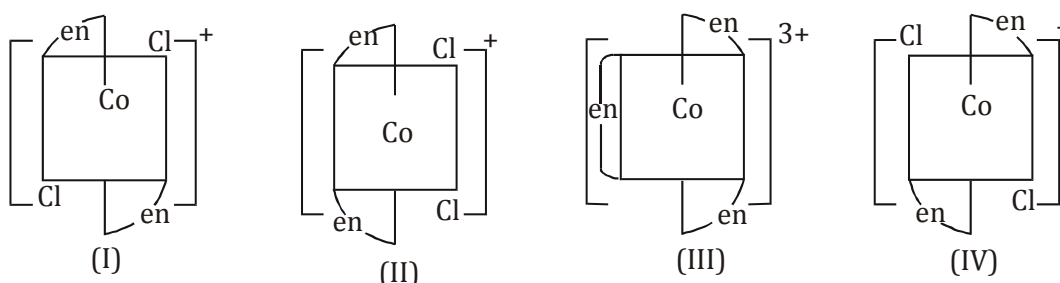
CCC238

86. Which of the following will show optical isomerism but not geometrical isomerism.

- (A) $[\text{Cr}(\text{Cl}_3)_3(\text{NH}_3)_3]$ (B) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$
 (C) $[\text{CoCl}_2(\text{en})_2]^{\oplus}$ (D) $[\text{CoBrCl}(\text{Ox})_2]^{3-}$

CCC242

87. Which of the following ions are optically active ?



- (A) I only (B) II only
 (C) II and III (D) IV only

CCC223

Stereoisomerism for Octahedral Complexes for Unsymmetrical Bidentate Ligand

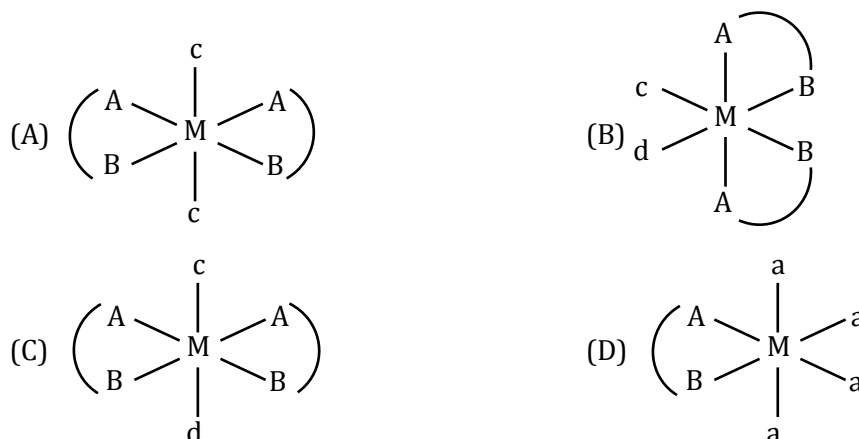
88. Find complex which have maximum number of stereoisomers -

- (A) $[\text{Ma}_6]$ (B) $[\text{Ma}_3\text{b}_3]$
 (C) $[\text{M}(\text{AB})\text{abcd}]$ (D) $[\text{M}(\text{AA})_2\text{b}_2]$

CCC225

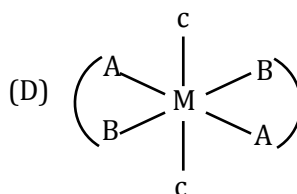
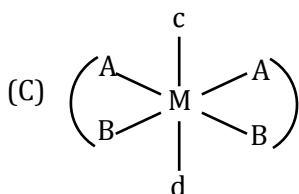
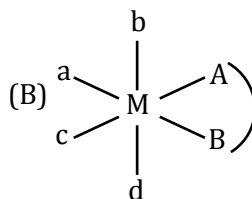
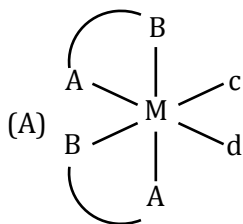
89. Which of the following octahedral complex is optically active ?

[M = Central atom, (a, b, c, d = monodentate ligands), (AB = unsymmetrical bidentate ligand)]



CCC227

90. Select the complex which has centre of symmetry : [M = central atom, AB = unsymmetrical bidentate ligand, (a, b, c, d = monodentate ligands)]



CCC228

91. The no of geometrical isomers for the complex $[MCl_2(L)_2]$ {where M = central atom, $L = NH_2 - CH_2COO^-$ } are :
 (A) 2 (B) 4 (C) 5 (D) 6

CCC230

Bonding in Coordination Compound; Valence Bond Theory; Questions of Valence Bond Theory; Limitation of Valence Bond Theory

92. Which of the following is the limitation of valence bond theory ?
 (A) It does not distinguish between weak and strong ligands.
 (B) It does not give quantitative interpretation of magnetic data.
 (C) It does not explain the colour exhibited by coordination compounds.
 (D) All of these

CCC231

Spectro chemical series

93. Which complex of Co^{2+} will have the weakest crystal field splitting -
 (A) $[CoCl_6]^{4-}$ (B) $[Co(CN)_6]^{4-}$ (C) $[Co(NH_3)_6]^{2+}$ (D) $[Co(en)_3]^{2+}$

CCC257

Crystal Field Theory for Octahedral Complexes

94. In an octahedral crystal field, the t_{2g} orbitals are
 (A) Raised in energy by $0.4 \Delta_o$ (B) Lowered in energy by $0.4 \Delta_o$
 (C) Raised in energy by $0.6 \Delta_o$ (D) Lowered in energy by $0.6 \Delta_o$
95. Consider the following complex $[Co(NH_3)_5(CO_3)]ClO_4$. The coordination number, oxidation number, number of d-electrons and number of unpaired d-electrons on the metal are respectively
 (A) 6, 3, 6, 0 (B) 7, 2, 7, 1 (C) 7, 1, 6, 4 (D) 6, 2, 7, 3
96. Which of the following facts about the complex $[Cr(NH_3)_6]Cl_3$ is wrong ?
 (A) The complex involves d^2sp^3 hybridisation and is octahedral in shape.
 (B) The complex is paramagnetic.
 (C) The complex is an outer orbital complex
 (D) The complex gives white precipitate with silver nitrate solution.

CCC266

CCC232

CCC234

Coordination Compounds

97. For which of the following types of d^n configuration, the number of unpaired electrons in octahedral complexes remains same irrespective of the ligand field strength.
(A) d^3 (B) d^4 (C) d^5 (D) d^6

CCC251

98. Which of the following compound is paramagnetic –
(A) Tetracyanonickelate(II) ion (B) Tetraamminezinc(II) ion
(C) Hexaminechromium(III) ion (D) Diamminesilver(I) ion

CCC252

Exceptions in Octahedral Complexes

99. The number of unpaired electrons calculated in $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{CoF}_6]^{3-}$ are :
(A) 4 and 4 (B) 0 and 2 (C) 2 and 4 (D) 0 and 4

CCC264

100. $[\text{M}(\text{C}_2\text{O}_4)_3]^{x-}$ is diamagnetic when M^{n+} ion is :
(A) Co^{3+} (B) Co^{2+} (C) Ni^{2+} (D) Fe^{2+}

CCC269

101. $[\text{NiF}_6]^{2-}$ is :
(A) Paramagnetic (B) high spin (C) Diamagnetic (D) tetrahedral

CCC268

102. Which of the following is correctly matched?
(A) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$: diamagnetic, $\Delta_o < \text{pairing energy}$
(B) $\text{K}_4[\text{Fe}(\text{CN})_6]$: Low spin complex, colourless
(C) $[\text{Co}(\text{H}_2\text{O})_6]\text{SO}_4$: Paramagnetic, high spin complex
(D) $[\text{NiF}_6]^{2-}$ diamagnetic, $\Delta_o < \text{pairing energy}$

CCC220

Exceptional Cases in Cr; Mn; Fe; Co; Ni Complexes

103. Which one of the following complex is an outer orbital complex -
(A) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ (B) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ (C) $[\text{Fe}(\text{NH}_3)_6]^{2+}$ (D) All of these

CCC256

104. The complex $[\text{Fe}(\text{NH}_3)_6]^{2+}$ is :
(A) Low spin (B) High spin
(C) Diamagnetic (D) both (A) and (C)

CCC267

Crystal Field Theory for Tetrahedral Complexes

105. The geometry of $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{PPh}_3)_2\text{Cl}_2]$ are
(A) both square planar (B) tetrahedral and square planar
(C) both tetrahedral (D) None of these

CCC233

106. Which of the following complex shows sp^3d^2 hybridization ?
(A) $[\text{Cr}(\text{NO}_2)_6]^{3-}$ (B) $[\text{Fe}(\text{CN})_6]^{4-}$ (C) $[\text{CoF}_6]^{3-}$ (D) $[\text{Ni}(\text{CO})_4]$

CCC236

107. What is the hybridisation of Fe in $[\text{Fe}(\text{CO})_4]^{2-}$
(A) d^2sp^3 (B) sp^3d^2 (C) sp^3d (D) sp^3

CCC247

108. Identify tetrahedral species which has maximum magnetic moment value :
(A) $[\text{CuCl}_4]^{2-}$ (B) $[\text{CoCl}_4]^{2-}$ (C) $[\text{FeCl}_4]^{2-}$ (D) $[\text{AlCl}_4]^-$

CCC249

- 109.** Tetrahedral crystal field splitting energy Δ_t is roughly how many times of octahedral crystal field splitting energy Δ_o ?
 (A) $1/2$ (B) $4/9$ (C) $9/4$ (D) 2 CCC271
- 110.** What CFSE of a tetrahedral complex with d^2 configuration of central atom ?
 (A) $-0.6 \Delta_t$ (B) $-1.2 \Delta_t$ (C) $-0.8 \Delta_t$ (D) 0 CCC276
- 111.** How many unpaired electrons are there in d^7 tetrahedral complex ?
 (A) 3 (B) 1 (C) 2 (D) 4 CCC277
- 112.** The value of crystal field stabilisation energy for a tetrahedral complex having d^5 configuration of central metal ion is :
 (A) $-0.6 \Delta_t$ (B) $-1.2 \Delta_t$ (C) $-0.4 \Delta_t$ (D) Zero CCC273

Calculation of Crystal Field Stabilisation Energy for Octahedral Complexes

- 113.** CFSE in d^3 , octahedral is :
 (A) $-1.2 \Delta_o$ (B) $-0.8 \Delta_o$
 (C) $+0.6 \Delta_o$ (D) $-2.4 \Delta_o$ CCC299
- 114.** If Δ_o is crystal field splitting energy and P is mean pairing energy per pair then stabilizing energy of complex $[\text{FeF}_6]^{4-}$ is -
 (A) $-2.4 \Delta_o + 3P$ (B) $-2.0 \Delta_o + 3P$
 (C) $-1.6 \Delta_o + P$ (D) $-0.4 \Delta_o$ CCC300

Crystal Field Theory for Square Planar Complexes

- 115.** The metal complex that is diamagnetic is (Atomic number: Fe = 26; Cu = 29)
 (A) $\text{K}_3[\text{Cu}(\text{CN})_4]$ (B) $\text{K}_2[\text{Cu}(\text{CN})_4]$ (C) $\text{K}_3[\text{Fe}(\text{CN})_4]$ (D) $\text{K}_4[\text{FeCl}_6]$ CCC235
- 116.** Which of the following complexes has square planar structure ?
 (A) $[\text{Ni}(\text{CO})_4]$ (B) $[\text{Ni}(\text{CN})_4]^{2-}$ (C) $[\text{Ni}(\text{Cl})_4]^{2-}$ (D) $[\text{Zn}(\text{NH}_3)_4]^{2+}$ CCC237
- 117.** Which of the following is diamagnetic and sp^3 hybridised -
 (A) $[\text{NiCl}_4]^{2-}$ (B) $[\text{Ni}(\text{CN})_4]^{4-}$ (C) $[\text{Ni}(\text{CN})_4]^{2-}$ (D) $[\text{NiCl}_2(\text{PPh}_3)_2]$ CCC248
- 118.** $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ do not differ in
 (A) magnetic moment (B) oxidation number of Ni
 (C) geometry (D) EAN CCC250
- 119.** The magnetic moment of $[\text{NiX}_4]^{2-}$ ion is found to be zero. Then the metal of the complex ion is (X = monodentate anionic ligand).
 (A) sp^3 hybridised (B) sp^2 hybridised (C) dsp^2 hybridised (D) d^2sp hybridised CCC255

Coordination Compounds

120. One unknown complex has the spin only magnetic moment is of 1.73 BM. As per the C. F. T., complex is.
 (A) d^7 , Oh-field, with Strong Field Ligand (B) d^9 , sq.planar-field, with Strong Field Ligand
 (C) d^9 , Td,field with Weak Field Ligand (D) All of these
 CCC274
121. How many unpaired electrons are there in square planar complex of Co^{2+} ?
 (A) 1 (B) 2 (C) 3 (D) 4
 CCC279
122. The number of unpaired electron in $[AuCl_4]^-$ is :
 (A) 0 (B) 1 (C) 2 (D) 4
 CCC280
123. $[PdCl_4]^{2-}$ is :
 (A) Tetrahedral, 2 unpaired e^- s. (B) Square planar, paramagnetic
 (C) Square planar, diamagnetic (D) Tetra hedral, diamagnetic
 CCC281
124. Which of the following is a correct statement?
 (A) $[Co(NH_3)_6]^{3+}$ is paramagnetic while $[CoF_6]^{3-}$ is diamagnetic
 (B) $[Pt(NH_3)(NH_2OH)(Py)(NO_2)]NO_3$ has square planar shape
 (C) $Ni(CO)_4$ possesses square planar geometry while $[Ni(CN)_4]^{2-}$ is tetrahedral.
 (D) IUPAC name for $[Co(NH_3)_4Cl(ONO)]Cl$ is tetraamminechloronitrocobalt (II) chloride.
 CCC311
125. The compound Na_2IrCl_6 reacts with triphenylphosphine in diethyleneglycol in an atmosphere of CO to give $[IrCl(CO)(PPh_3)_2]$, known as 'Vaska's compound'. (Atomic number of Ir = 77) Which of the following statements is correct?
 (A) The IUPAC name of the complex is carbonylchloridobis (triphenylphosphine) iridium(III)
 (B) The hybridisation of the metal ion is sp^3
 (C) The magnetic moment (spin only) of the complex is zero
 (D) The complex shows geometrical as well as ionization isomerism
 CCC289

Factors Affecting Crystal Field Stabilisation Energy

126. Crystal field stabilisation energy of a complex is affected by :
 (A) Ligand strength (B) Oxidation Number of central atom
 (C) Coordination number (D) All of these
 CCC296
127. Δ_o is maximum for which of the following given complex?
 (A) $[Co(H_2O)_3F_3]$ (B) $[Co(NH_3)_6]^{3+}$
 (C) $[Co(H_2O)_6]^{3+}$ (D) $[Co(NH_3)_4]^{3+}$
 CCC297
128. Select the correct general order of comparison :
 (A) $\Delta_t > \Delta_{sp} > \Delta_o$ (B) $\Delta_{sp} > \Delta_o > \Delta_t$
 (C) $\Delta_{sp} > \Delta_t > \Delta_o$ (D) $\Delta_o > \Delta_{sp} > \Delta_t$
 CCC298
129. Among the following species the one which causes the highest CFSE, Δ_o as a ligand is :-
 (A) CN^- (B) NH_3 (C) CO (D) F^-
 CCC301

Colour in Transition Metal Coordination Compound Using $d-d$ Transition; Charge Transfer Spectrum; Complementary Colour Wheel;

130. $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ ion is :
 (A) colourless and diamagnetic (B) coloured and octahedral
 (C) colourless and paramagnetic (D) coloured and paramagnetic
 CCC291
131. An ion M^{2+} , forms the complexes $[\text{M}(\text{H}_2\text{O})_6]^{2+}$, $[\text{M}(\text{en})_3]^{2+}$ and $[\text{MBr}_6]^{4-}$, match the complex with the appropriate colour.
 (A) Green, blue and red (B) Blue, red and green
 (C) Green, red and blue (D) Red, blue and green
 CCC302
132. If $\lambda_{\text{absorbed}}$ for $d-d$ transition is in order $[\text{Ti}(\text{X})_6]^{3+} > [\text{Ti}(\text{Y})_6]^{3+} > [\text{Ti}(\text{Z})_6]^{3+}$.
 Select correct order of strength of ligands (X, Y, Z are monodentate ligand)-
 (A) $\text{Z} > \text{Y} > \text{X}$ (B) $\text{X} > \text{Y} > \text{Z}$
 (C) $\text{Z} > \text{X} > \text{Y}$ (D) Not predictable
 CCC303
133. Which of the following complex is coloured and diamagnetic –
 (A) MnO_4^{2-} (B) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
 (C) $[\text{CrCl}_6]^{3-}$ (D) CrO_4^{2-}
 CCC304
134. Which of the following compound is not coloured ?
 (A) $\text{Na}_2[\text{CuCl}_4]$ (B) $\text{Na}_2[\text{CdCl}_4]$ (C) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (D) $\text{K}_3[\text{Fe}(\text{CN})_6]$
 CCC307
135. Given the following data about the absorption maxima of several complex ions, what is the order of Δ_0 for these ions.
- | Compound | λ_{max} |
|---|------------------------|
| I. $[\text{CrCl}_6]^{3-}$ | 758 |
| II. $[\text{Cr}(\text{NH}_3)_6]^{3+}$ | 465 |
| III. $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ | 694 |
- (A) $\Delta_0 (\text{I}) < \Delta_0 (\text{II}) < \Delta_0 (\text{III})$ (B) $\Delta_0 (\text{II}) < \Delta_0 (\text{III}) < \Delta_0 (\text{I})$
 (C) $\Delta_0 (\text{I}) < \Delta_0 (\text{III}) < \Delta_0 (\text{II})$ (D) $\Delta_0 (\text{III}) < \Delta_0 (\text{II}) < \Delta_0 (\text{I})$
 CCC308

Synergic Bonding

136. Statement-1 : In Zeise's salt C – C bond order regarding ethylene molecule is less than two.
 Statement-2 : Ethylene accepts electron pair from filled d -orbital of Pt^{2+} into its vacant bonding M.O.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.
 CCC120
137. Which of the following complex has highest M – C bond order ?
 (A) $[\text{Mn}(\text{CO})_6]^+$ (B) $[\text{V}(\text{CO})_6]^+$ (C) $[\text{Cr}(\text{CO})_6]$ (D) $[\text{Ti}(\text{CO})_6]^{2-}$
 CCC135

138. In the isoelectronic series of metal carbonyl, the C–O bond strength is expected to increase in the order.
 (A) $[\text{Mn}(\text{CO})_6]^{\oplus} < [\text{Cr}(\text{CO})_6] < [\text{V}(\text{CO})_6]^{\ominus}$
 (B) $[\text{V}(\text{CO})_6]^{\ominus} < [\text{Cr}(\text{CO})_6] < [\text{Mn}(\text{CO})_6]^{\oplus}$
 (C) $[\text{V}(\text{CO})_6]^{\ominus} < [\text{Mn}(\text{CO})_6]^{\oplus} < [\text{Cr}(\text{CO})_6]$
 (D) $[\text{Cr}(\text{CO})_6] < [\text{Mn}(\text{CO})_6]^{\oplus} < [\text{V}(\text{CO})_6]^{\ominus}$

CCC136

139. Which of the following order is correct for the IR vibrational frequency of CO.
 (A) $[\text{Fe}(\text{CO})_4]^{2\ominus} < [\text{Co}(\text{CO})_4]^{\ominus} < [\text{Ni}(\text{CO})_4]$
 (B) $[\text{Fe}(\text{CO})_4]^{2\ominus} > [\text{Co}(\text{CO})_4]^{\ominus} > [\text{Ni}(\text{CO})_4]$
 (C) $[\text{Fe}(\text{CO})_4]^{2\ominus} > [\text{Co}(\text{CO})_4]^{\ominus} < [\text{Ni}(\text{CO})_4]$
 (D) $[\text{Fe}(\text{CO})_4]^{2\ominus} < [\text{Co}(\text{CO})_4]^{\ominus} > [\text{Ni}(\text{CO})_4]$

CCC137

140. Which of the following complex has highest C–O bond length ?
 (A) $[\text{Ni}(\text{CO})_3(\text{PF}_3)]$ (B) $[\text{Ni}(\text{CO})_4]$
 (C) $[\text{Ni}(\text{CO})_3(\text{PMe}_3)]$ (D) None of these

CCC138

141. Metal-ligand bond order is more than 1 in which of the following complex species ?
 (A) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (B) $[\text{Cr}(\text{CN})_6]^{3\ominus}$ (C) $[\text{V}(\text{CO})_6]^{\ominus}$ (D) $[\text{Sc}(\text{H}_2\text{O})_6]\text{Cl}_3$

CCC139

142. Select the incorrect statement about metal carbonyl complex compounds :
 (A) Metal carbon bond in metal carbonyls possess both σ and π character
 (B) Due to synergic bonding metal carbon bond becomes weak
 (C) Due to synergic bonding carbon oxygen bond strength decreases
 (D) In metal carbonyls extent of synergic bonding will increase with increase in negative charge on central metal ion

CCC140

Importance and Applications of Coordination Compounds

143. Which one of the following coordination compounds is used to inhibit the growth of tumours ?
 (A) Trans-platin (B) EDTA complex of calcium
 (C) $[(\text{Ph}_3\text{P})_3\text{RhCl}]$ (D) Cis-platin

CCC283

144. The compound used in the treatment of lead poisoning is :
 (A) D-penicillamine (B) desferrioxime B
 (C) Cis-platin (D) EDTA

CCC284

145. Coordination compounds have great importance in biological system. In this context which of the following statements is incorrect.
 (A) Cyanocobalamin is B_{12} and contains cobalt
 (B) Haemoglobin is the red pigment of blood and contains iron
 (C) Chlorophylls are green pigments in plants and contain calcium
 (D) Carboxypeptidase – A is an enzyme and contains zinc.

CCC285

146. Which of the following is not considered as an organometallic compound ?
 (A) Ferrocene (B) Cis-platin (C) Ziese's salt (D) Grignard reagent

CCC121

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Introduction of Coordination Compounds

1. Which is TRUE about metal complex ?
 (A) Central metal acts as lewis acid
 (B) Ligands acts lewis base
 (C) It will give test of all individual entities in aqueous solution
 (D) The coordinate bonds are present in the metal complex

CCC108

Types of Complex: Homoleptic and Heteroleptic Complex

2. Which of the following complexes is/are an example of homoleptic complex ?
 (A) $[\text{PtCl}_4]^{2-}$ (B) $\text{K}_3[\text{Fe}(\text{CN})_6]$
 (C) $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$ (D) $[\text{CoBr}_2(\text{en})_2]^+$

CCC109

3. Which of the following complexes is/are an example of heteroleptic complex ?
 (A) $[\text{Co}(\text{NO}_2)_3(\text{NH}_3)_3]$ (B) $\text{K}_3[\text{CrCl}_6]$
 (C) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (D) $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$

CCC110

Types of Ligands: On the basis of Denticity; Ligand as Polydentate Ligand; Chelating Ligands; Ambidentate Ligands; Flexidentate Ligands;

4. Which of the following bidentate ligand(s) has similar donor atoms :
 (A) py (B) bn (C) en (D) gly⁻

CCC129

Classical and Non Classical Ligands

5. Which of the following chemical species can act as non-classical ligand(s)
 (A) CO (B) C_2H_4 (C) NO^+ (D) PR_3

CCC130

Effective Atomic Number; Calculation of EAN; Sidwick EAN Rule; Effective Atomic Number of Bridging Complexes; Oxidising and Reducing Property of Carbonyl Complex

6. Which of the following complex is following sidwick EAN rule.
 (A) $[\text{Ni}(\text{CO})_4]$ (B) $\text{K}_2[\text{Ni}(\text{CN})_4]$ (C) $\text{K}_2[\text{HgI}_4]$ (D) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$

CCC168

7. Which of the following pairs of compounds has the same EAN value but does not obey Sidwick EAN rule?
 (A) $\text{Fe}(\text{CO})_5$, $\text{Ni}(\text{CO})_4$ (B) $[\text{Fe}(\text{NH}_3)_6]^{2+}$, $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$
 (C) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$, $[\text{Mn}(\text{CO})_6]$ (D) $[\text{Ti}(\text{CO})_6]$, $[\text{Mn}(\text{C}_2\text{O}_4)_3]^{3-}$

CCC324

Stereoisomerism for Square Planar Complexes

8. Select which Square planar complex (es) cannot show optical isomerism : -
 (A) $[\text{Pt}(\text{bn})_2]^{2+}$ (B) $[\text{Pt}(\text{en})_2]^{2+}$ (C) $[\text{PtCl}_2(\text{NH}_3)_2]^0$ (D) $[\text{Pt}(\text{gly})_2]^0$

CCC212

Stereoisomerism for Octahedral Complexes for Monodentate Ligand

9. Which of the following exhibit geometrical isomerism (M stands for a metal, and a and b are achiral ligands) ?
 (A) Ma_2b_2 (Sq. Pl.) (B) Ma_4b_2 (C) Ma_5b (D) Ma_6

CCC240

Stereoisomerism for Octahedral Complexes for Symmetrical Bidentate Ligand

10. Which of the following will have two stereoisomeric forms?
 (A) $[\text{Cr}(\text{NO}_3)_3(\text{NH}_3)_3]$ (B) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$
 (C) $[\text{CoCl}_2(\text{en})_2]^{\oplus}$ (D) $[\text{CoBrCl}(\text{Ox})_2]^{3\ominus}$

CCC241

Exceptions in Octahedral Complexes

11. $[\text{MnCl}_6]^{3\ominus}$, $[\text{FeF}_6]^{3\ominus}$ and $[\text{CoF}_6]^{3\ominus}$ complexes are -
 (A) Outerorbital
 (B) Having sp^3d^2 hybridisation
 (C) Paramagnetic in nature
 (D) Having four, five and four unpaired electrons respectively

CCC262

Exceptional Cases in Cr; Mn; Fe; Co; Ni Complexes

12. In case of $[\text{Mn}(\text{NH}_3)_6]^{2+}$
 (A) relation in between Δ_0 and P is $\Delta_0 < P$.
 (B) two unpaired electrons are present in axial d-orbital of Mn^{2+} .
 (C) hybridization of central metal atom of complex is d^2sp^3 .
 (D) d-orbitals involved in hybridization of central atom are d_{x^2} and d_{xy} .

CCC322

Crystal Field Theory for Tetrahedral Complexes

13. Which of the following ion is/are diamagnetic and non planar ?
 (A) $[\text{Ni}(\text{CN})_4]^{2\ominus}$ (B) $\text{MnO}_4^{\ominus}$
 (C) $[\text{Cu}(\text{NH}_3)_4]^{2\oplus}$ (D) $\text{CrO}_4^{2\ominus}$

CCC315

Crystal Field Theory for Square Planar Complexes

14. Select pair(s) in which given complex is isostructural with given xenon fluoride.
 (A) XeF_2 ; $[\text{AgCl}_2]^{\ominus}$ (B) XeF_4 ; $[\text{NiCl}_4]^{2\ominus}$ (C) XeF_4 ; $[\text{PtCl}_4]^{2\ominus}$ (D) XeF_6 ; $[\text{PtCl}_4]^{2\ominus}$
15. Which of the following statement(s) is/are correct?
 (A) $\text{Ni}(\text{CO})_4$ is diamagnetic and sp^3 hybridized.
 (B) Complex $[\text{Cu}(\text{CN})_4]^{3-}$ is paramagnetic and Cu^{2+} ion is dsp^2 hybridized.
 (C) $[\text{MnBr}_4]^{2-}$ ion is paramagnetic and sp^3 hybridized.
 (D) Vaska's catalyst, that is, $[\text{IrCl}(\text{CO})(\text{PPh}_3)_2]$ is diamagnetic and Ir^+ ion is dsp^2 hybridized.
16. Which of the following complexes is/are square planar?
 (A) $[\text{AgF}_4]^-$ (B) $[\text{AuCl}_4]^-$ (C) $[\text{NiCl}_2(\text{PPh}_3)_2]$ (D) $[\text{NiCl}_2(\text{Pme}_3)_2]$

CCC263

CCC321

CCC323

17. In which of the following complex(s) spin only magnetic moment is independent, from the nature of ligand. (L = monodentate ligand) -
 (A) $[\text{NiL}_4]^{2\oplus}$ (B) $[\text{NiL}_6]^{2\oplus}$ (C) $[\text{FeL}_6]^{3\oplus}$ (D) $[\text{CrL}_6]^{3\oplus}$

CCC314

Factors Affecting Crystal Field Stabilisation Energy

18. Select the correct stability order for the following complexes.
 (A) $[\text{Cu}(\text{trien})]^{2\oplus} > [\text{Cu}(\text{en})_2]^{2\oplus} > [\text{Cu}(\text{NH}_3)_4]^{2\oplus}$
 (B) $[\text{Ir}(\text{H}_2\text{O})_6]^{3\oplus} > [\text{Rh}(\text{H}_2\text{O})_6]^{3\oplus} > [\text{Co}(\text{H}_2\text{O})_6]^{3\oplus}$
 (C) $[\text{Cr}(\text{NH}_3)_6]^{3\oplus} > [\text{Cr}(\text{NH}_3)_6]^{2\oplus} > [\text{Cr}(\text{NH}_3)_6]^\oplus$
 (D) $[\text{Cu}(\text{CO})_4]^{2\oplus} > [\text{Cu}(\text{H}_2\text{O})_4]^{2\oplus} > [\text{Cu}(\text{NH}_3)_4]^{2\oplus}$

CCC313

METRIX MATCH TYPE QUESTIONS

Types of Ligands: On the basis of Denticity; Ligand as Polydentate Ligand; Chelating Ligands; Ambidentate Ligands; Flexidentate Ligands;

19. Match each coordination compound in List-I with co-ordination number of central metal/ion from List-II and select the correct answer using the code given below the lists.

List-I

- (P) $[\text{Co}(\text{en})_3]^{3\oplus}$
 (Q) $[\text{Ca}(\text{EDTA})]^{2\ominus}$
 (R) $[\text{Ni}(\text{CO})_4]$
 (S) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
 (A) P → (1), Q → (3), R → (2), S → (3)
 (C) P → (2), Q → (1), R → (2), S → (2)

List-II

- (1) 6
 (2) 4
 (3) 2
 (4) 5
 (B) P → (1), Q → (1), R → (2), S → (3)
 (D) P → (2), Q → (2), R → (3), S → (2)

CCC127

20. Match the List with multiple options

List-I

- (P) Ethane 1, 2 diamine
 (Q) $\text{SCN}^\ominus$
 (R) $\text{CO}_3^{2\ominus}$
 (S) $\text{NO}_2^\ominus$
 Code :

P	Q	R	S
(A) 1	2	3	4
(C) 4	1	3	2

List-II

- (1) Chelating agent
 (2) Ambidentate
 (3) Flexidentate
 (4) Tridentate

P	Q	R	S
(B) 1	2	3	2
(D) 4	1	2	4

CCC128

Werner Theory; Electrical and Molar Conductivity

21. Match List I with List II and select the correct answer using the codes given below the lists :-

**List I (Equivalent)
(conductance)**

- A. 229
 B. 97
 C. 404
 D. 523
 (A) A→i; B→iii; C→ii; D→iv
 (C) A→i; B→iii; C→iv; D→ii

List II

(Formula)

- i. $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$
 ii. $[\text{Pt}(\text{NH}_3)_3\text{Cl}_3]\text{Cl}$
 iii. $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$
 iv. $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$
 (B) A→i; B→iv; C→iii; D→ii
 (D) A→iii; B→ii; C→i; D→iv

CCC152

Effective Atomic Number; Calculation of EAN; Sidwick EAN Rule; Effective Atomic Number of Bridging Complexes; Oxidising and Reducing Property of Carbonyl Complex

22. Match the complexes in column-I with the EAN of central atom in column-II:

Column-I

- (A) $[\text{Fe}(\text{CO})_4]^{2\ominus}$
 (B) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 (C) $\text{K}_2[\text{Ni}(\text{CN})_4]$
 (D) $[\text{Cu}(\text{NH}_3)_4]^{2\oplus}$
 (A) A-(P), B-(R), C-(P), D-(S)
 (C) A-(P), B-(S), C-(R), D-(Q)

Column-II

- (P) 34
 (Q) 35
 (R) 36
 (S) 37
 (B) A-(R), B-(R), C-(P), D-(Q)
 (D) A-(R), B-(S), C-(P), D-(Q)

CCC163

Stereoisomerism for Octahedral Complexes for Symmetrical Bidentate Ligand

23. Match the complexes in column I with their stereoproperties in column II

Column I

- (A) $[\text{CoCl}_3(\text{NH}_3)_3]$
 (B) $[\text{Cr}(\text{ox})_3]^{3\ominus}$
 (C) $[\text{CrCl}_2(\text{ox})_2]$
 (D) $[\text{RhCl}_3(\text{Py})_3]$
 (A) A-P, R B-T, C-Q, S, T D-P, Q
 (C) A-P, S B-T, C-Q, R, T D-P, S

Column II

- (P) Has a facial isomer
 (Q) Cis form is optically active
 (R) Trans form is optically inactive
 (S) Has a meridional form
 (T) Two optically active isomer
 (B) A-P, S B-T, C-Q, S, T D-P, S
 (D) A-P, R B-T, C-Q, R, T D-P, Q

CCC239

Crystal Field Theory for Octahedral Complexes

24. **Column - I**

- (A) $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$
 (B) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$
 (C) $[\text{Ag}(\text{CN})_2]^\ominus$
 (D) $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (A) (A-P, Q, R B-Q, R, T C-P D-P, R)
 (C) (A-P, Q, S B-Q, R, T C-Q D-P, Q)

Column -II

- (P) $\mu = 0$ B.M.
 (Q) octahedral
 (R) $\mu = \sqrt{15}$ B.M.
 (S) $\text{NO}^\oplus$ ligand
 (B) (A-P, Q, S B-Q, R, S C-P D-P, Q)
 (D) (A-P, Q, R B-Q, R, S C-P D-P, S)

CCC258

25. Match List-I (Complex ions) with List-II (Number of Unpaired Electrons) and select the correct answer using the codes given below the lists :-

List -I

(Complex ions)

- A. $[\text{CrF}_6]^{4\ominus}$
 B. $[\text{MnF}_6]^{4\ominus}$
 C. $[\text{Cr}(\text{CN})_6]^{4\ominus}$
 D. $[\text{Mn}(\text{CN})_6]^{4\ominus}$

List II

(Number of Unpaired Electrons)

- i. One
 ii. Two
 iii. Three
 iv. Four
 v. Five

Code :

- A B C D**
 (A) iv i ii v
 (C) iv v ii I

- A B C D**
 (B) ii v iii i
 (D) ii i iii v

CCC272

Crystal Field Theory for Square Planar Complexes

Answer Q.26, Q.27 and Q.28 by appropriately matching the information given in the three columns of the following table.

For the following complexes in Column-I, match the correct properties with Column-II and Column-III according to the questions asked.

Column-I	Column-II Hybridisation	Column-III Magnetic moment value in B.M.
(I) $[\text{Co}(\text{H}_2\text{O})_3\text{F}_3]$	(i) sp^3	(p) 2.8
(II) $[\text{NiF}_6]^{2\ominus}$	(ii) dsp^2	(q) 5.92
(III) $\text{Na}_2[\text{NiCl}_4]$	(iii) sp^3d^2	(r) 0
(IV) $\text{K}_2[\text{Ni}(\text{CN})_4]$	(iv) d^2sp^3	(s) 4.9

26. Which is the only correct combination?
 (A) I, (iv), (r) (B) II, (i), (q) (C) IV, (ii), (p) (D) IV, (ii), (r) CCC259
27. Which is the only incorrect combination ?
 (A) IV, (ii), (r) (B) II, (iii), (p) (C) II, (iv), (r) (D) III, (i), (p) CCC260
28. Which is the only correct combination for $[\text{Co}(\text{H}_2\text{O})_3\text{F}_3]$?
 (A) (iv), (r) (B) (iii), (p) (C) (iii), (s) (D) (iv), (q) CCC261

Synergic Bonding

29. Match the list :

List-I

[Species]

- (P) CO
 (Q) $\text{Cr}(\text{CO})_6$
 (R) $[\text{V}(\text{CO})_6]^\ominus$
 (S) $[\text{Mn}(\text{CO})_6]^\oplus$
 (A) P→1; Q→2; R→3; S→4
 (C) P→3; Q→1; R→2; S→4

List-II

[Stretching frequency of CO (cm^{-1})]

- (1) 2143
 (2) 2000
 (3) 1860
 (4) 2090
 (B) P→2; Q→1; R→3; S→4
 (D) P→4; Q→1; R→3; S→2

CCC151

Importance and Applications of Coordination Compounds

30. Match the List-I with List-II :

List-I

- (P) Ferrocene
 (Q) $\text{Mn}_2(\text{CO})_{10}$
 (R) Vitamine B₁₂
 (S) Haemoglobin

List-II

- (1) Iron present
 (2) Cobalt
 (3) Metal-Metal bonding
 (4) Sandwich structure

Code :

- | | P | Q | R | S |
|-----|-----|---|---|---|
| (A) | 4 | 3 | 1 | 2 |
| (B) | 1 | 3 | 1 | 2 |
| (C) | 1,4 | 3 | 2 | 1 |
| (D) | 1 | 3 | 4 | 2 |

CCC309

Table for Question 31 to 33

Complex		Oxidation state of central metal atom or ion		Ligand (s) present	
(1)	Ferrocene	(P)	0	(i)	$C_5H_5^\ominus$
(2)	Potassium ferrocynide	(Q)	+1	(ii)	$NO^\oplus$
(3)	Sodium nitroprusside	(R)	+2	(iii)	$CN^\ominus$
(4)	Zeise's salt	(S)	+3	(iv)	$Cl^\ominus$

31. Which of the following matching is INCORRECT : -

- (A) (1),(R),(i) (B) (2), (R), (iii)
(C) (3), (R), (ii) and (iii) (D) (4), (S), (iv)

CCC189

32. Select INCORRECT combination for complex given : -

- (A) Complex having platinum = R, (iv)
(B) Neutral / uncharged complex = R, (i)
(C) Complex which have η^2 - bonded ligand = R, (i)
(D) Sandwich complex = R, (i)

CCC190

33. Which of the following complex having π -acid ligand (s) : -

- (A) 1,2,3,4 (B) 1, 2, 3 (C) 1, 2 (D) 1, 4

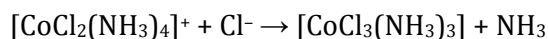
CCC191

COMPREHENSION TYPE QUESTIONS

Stereoisomerism for Octahedral Complexes for Monodentate Ligand

Paragraph for Question No. 34 to 36

Coordination number of a central atom is its inherent property and has no role on its isomerism.



34. If in the above reaction two isomers of the product are obtained, which is true for the initial (reactant) complex?

- (A) compound is in cis form.
(B) compound is in trans form.
(C) compound is in both (cis and trans) forms.
(D) cannot be predicted.

CCC328

35. When 0.1 mol of $CoCl_3(NH_3)_5$ is treated with excess of $AgNO_3$, 0.2 mol of $AgCl$ are obtained. The conductivity of solution will correspond to (cation : anion)

- (A) 1 : 3 electrolyte (B) 1 : 2 electrolyte
(C) 1 : 1 electrolyte (D) 3 : 1 electrolyte

CCC329

36. Due to the presence of ambidentate ligands coordination compounds show isomerism, palladium complexes of the type $[Pd(C_6H_5)_2(SCN)_2]$ and $[Pd(C_6H_5)_2(NCS)_2]$ are

- (A) linkage isomers. (B) coordination isomers.
(C) ionisation isomers. (D) geometrical isomers.

CCC330

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Introduction of Coordination Compounds

- Find the number of correct statement(s) is/are :-
 - Both double salts as well as complexes are formed by the combination of two or more stable compounds in stoichiometric ratio
 - double salts such as carnallite, $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$, potash alum, $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$, etc. dissociate into simple ions completely when dissolved in water
 - complex ions such as $[\text{Fe}(\text{CN})_6]^{4-}$ or $\text{K}_4[\text{Fe}(\text{CN})_6]$, do not dissociate into Fe^{2+} and CN^- ions
 - Mohr's salt, $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ does not dissociate into simple ions completely when dissolved in water
 - $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$ complex dissociate into Co^{3+} and Cl^- ions

CCC111

Types of Ligands: On the basis of Denticity; Ligand as Polydentate Ligand; Chelating Ligands; Ambidentate Ligands; Flexidentate Ligands;

- What is the sum of coordination number of the central metal ions in the following complexes

(i) $[\text{Cu}(\text{NH}_3)_4]^{2+}$	(ii) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$	(iii) $[\text{PtCl}_2(\text{en})_2]^{2+}$
(iv) $[\text{Mo}(\text{CN})_6]^{4-}$	(v) $[\text{Fe}(\text{EDTA})]^-$	(vi) $[\text{PdI}_2(\text{ONO})_2(\text{H}_2\text{O})_2]$

CCC112

- Find the total number of 5-membered rings present in $[\text{Co}(\text{EDTA})]^-$.

CCC131

- Find the number of complex(es) which have at least one five member chelate ring formed by two carbon atoms, two nitrogen atoms and one central metal.

(I) $[\text{Co}(\text{EDTA})]^-$	(II) $[\text{Co}(\text{en})_3]^{3+}$	(III) $[\text{Co}(\text{Gly})_3]^0$	(IV) $[\text{Co}(\text{bipy})_3]^{3+}$
(V) $[\text{Co}(\text{oxalate})_3]^{3-}$	(VI) $[\text{Co}(\text{dien})(\text{NH}_3)_3]^{3+}$		

CCC132

Classical and Non Classical Ligands

- Find the number of ligands which behave predominantly as π -donor ligand (as well as non-classical ligand)

C_2H_4 , C_6H_6 , NO^+ , NO , CO , NH_3 , NH_2^- , H_2O , OH^-

CCC133

Effective Atomic Number; Calculation of EAN; Sidwick EAN Rule; Effective Atomic Number of Bridging Complexes; Oxidising and Reducing Property of Carbonyl Complex

- Determine the effective atomic number of the metal atom in $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$

CCC170

- Find the EAN value of central atom of $[\text{Fe}(\pi\text{-C}_4\text{H}_4)(\text{CO})_3]$

CCC171

Structural Isomerism

8. The number of coordination isomers possible for:

- (a) $[\text{Pt}(\text{NH}_3)_4 \text{Cl}][\text{PtCl}_4]$ is _____.
 (b) $[\text{Pt}(\text{NH}_3)_4 \text{Cl}_2][\text{Pt}(\text{CN})_4]$ is _____.
 (c) $[\text{Fe}(\text{NH}_3)_6]^{3+} [\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$ is _____.
 (d) $[\text{Pt}(\text{NH}_3)_4]^{2+} [\text{CuCl}_4]^{2-}$ is _____.
 (e) $[\text{Pt}(\text{NH}_3)_3 \text{Cl}]_2 [\text{Pt}(\text{SCN})_4]$ is _____.

CCC331

Stereoisomerism for Square Planar Complexes

9. Total number of geometrical isomers for the complex $[\text{RhCl}(\text{CO})(\text{PPh}_3)(\text{NH}_3)]$ is.

CCC213

10. Write all the geometrical isomers of $[\text{Pt}(\text{NH}_3)(\text{Br})(\text{Cl})(\text{py})]$ and how many of these will exhibit optical isomers?

CCC214

Stereoisomerism for Octahedral Complexes for Symmetrical Bidentate Ligand

11. Find the number of optically active isomers for $[\text{Pd}(\text{en})_2(\text{NH}_3)(\text{H}_2\text{O})]^{4+}$ cation.

CCC244

12. Number of possible Geometrical Isomers of $[\text{PtFClBrI}(\text{H}_2\text{O})(\text{NH}_3)]$

CCC246

13. Determine total number of optically active isomers of $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$.

CCC245

Crystal Field Theory for Octahedral Complexes

14. Find the maximum number of atoms lying in one plane for $[\text{Cr}(\text{CN})_6]^{3-}$

CCC243

Exceptions in Octahedral Complexes

15. The number of unpaired electrons in the t_{2g} set of d orbitals in the complex $[\text{Co}(\text{H}_2\text{O})_3\text{F}_3]$ is ____.

CCC333

Calculation of Crystal Field Stabilisation Energy for Octahedral Complexes

16. If crystal field stabilization energy of $[\text{ML}_6]^{n+}$ is $-0.8 \Delta_0$.

Find minimum number of electrons in t_{2g} orbitals of metal ion?

CCC294

Crystal Field Theory for Square Planar Complexes

17. Number of unpaired electrons in $\text{K}_3[\text{Fe}(\text{CN})_6]$, $\text{K}_2[\text{NiCl}_4]$, $\text{K}_2[\text{CoCl}_4]$, $\text{Na}_2[\text{Ni}(\text{CN})_4]$ are x, y, z and w respectively find $x + y + z + w = ?$

CCC293

18. Total number of complex compound which have low spin as well as Diamagnetic.

$[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{NiF}_6]^{2-}$, $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$,
 $[\text{Ni}(\text{en})_3]^{2+}$, $[\text{PtCl}_4]^{2-}$

CCC295

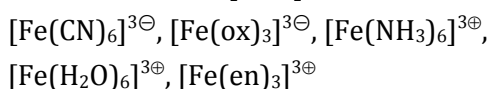
Factors Affecting Crystal Field Stabilisation Energy

19. Suppose we replace all the fluoride and water ligands from $[\text{CoF}_3(\text{H}_2\text{O})_3]$ by oxalate ligand without changing the oxidation state and coordination number of central metal atom or ion. Predict the number of parameters among following which increase in the newly formed complex.

- (I) Δ_0
 (II) Number of t_{2g} electrons
 (III) Number of e_g electrons
 (IV) Number of stereoisomers
 (V) Number of geometrical isomers
 (VI) EAN value

CCC332

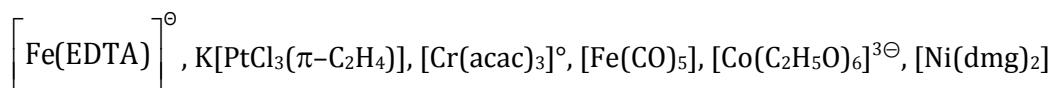
20. Find the number of complexes in which stability constant value is greater than the stability constant value of $[\text{FeF}_6]^{3-}$.



CCC334

Synergic Bonding

21. Select complex in which metal-carbon linkage(s) is / are present :



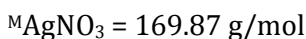
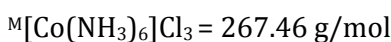
CCC153

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Werner Theory; Electrical and Molar Conductivity

22. The volume (in mL) of 0.125 M AgNO_3 required to quantitatively precipitate chloride ions in 0.3 g of $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ is _____.



CCC336

Effective Atomic Number; Calculation of EAN; Sidwick EAN Rule; Effective Atomic Number of Bridging Complexes; Oxidising and Reducing Property of Carbonyl Complex

23. Consider a metal complex ion $[\text{M}(\text{CO})_5]$. If the EAN (effective atomic number) of metal in complex is 36, then find out the following ratio for central metal atom (M) in the isolated state

$$\frac{\text{number of d electrons}}{\text{sum of number of s \& p electrons}}$$

CCC337

Stereoisomerism for Square Planar Complexes

24. Consider the given complex ion $[\text{Pt}(\text{bn})_2]^{2\oplus}$, where bn stands for butylene diamine. Find the value of $\frac{n}{2}$ where n is the total number of stereoisomers.

CCC338

Stereoisomerism for Octahedral Complexes for Monodentate Ligand

25. What is the number of total stereoisomers for the complex of type $[\text{Ma}_2\text{b}_2\text{cd}]$.

CCC339

Stereoisomerism for Octahedral Complexes for Symmetrical Bidentate Ligand

26. Find out the ratio of optically active to optically inactive compounds among the following
 $[\text{Be}(\text{bcac})_2]$; $[\text{CoBrCl}(\text{NO}_2)_4]^{3\ominus}$
 $[\text{Zn}(\text{gly})_2]$; $[\text{Pt}(\text{en})\text{Cl}_2]$;
 $[\text{CoF}_3(\text{H}_2\text{O})_3]$; $[\text{Co}(\text{EDTA})]^\ominus$
 $[\text{Pt}(\text{en})_3]^{4\oplus}$

CCC340

Crystal Field Theory for Octahedral Complexes

27. For the $t_{2g}^6 e_g^2$ system, the value of magnetic moment (μ) is :-

CCC341

28. For the co-ordination compound, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Br}$
 Total number of axial d-orbitals used for hybridisation in central atom = x
 Total number of stereoisomers = p
 Maximum number of atoms that may lie in same plane = q
 The value of $\left(\frac{q}{p \cdot x}\right)$ is-

CCC342

Calculation of Crystal Field Stabilisation Energy for Octahedral Complexes

29. For an octahedral complex of $\text{Fe}^{3\oplus}$ the value of C.F.S.E. (Crystal field stabilization energy) is " $x\Delta_0$ " if magnetic moment of complex is $\sqrt{35}$ B.M. Find out the value of |x|. (Δ_0 is crystal field splitting energy in octahedral field)

CCC343

Crystal Field Theory for Square Planar Complexes

30. Consider the following complex compounds.
 $[\text{CoCl}_2\text{Br}_2]^{2\ominus}$, $[\text{Ni}(\text{en})_3]^{2\oplus}$, $[\text{Ti}(\text{H}_2\text{O})_6]^{3\oplus}$, $[\text{Fe}(\text{NH}_3)_6]^{2\oplus}$, $[\text{Co}(\text{H}_2\text{O})_3\text{F}_3]$, $\text{K}_3[\text{Fe}(\text{CN})_6]$
 If number of low spin complexes is "X" and number of high spin complexes is "Y" then $\frac{X}{Y}$ is ____.

CCC344

EXERCISE - JEE (Main) PYQ

- The complex of Ni^{2+} ion and dimethyl glyoxime contains ____ number of Hydrogen (H) atoms.
[JEE (Main) 2025]
CCC382
- The set which does not have ambidentate ligand(s) is
[JEE (Main) 2023]
(1) $\text{C}_2\text{O}_4^{2-}$, ethylene diamine, H_2O (2) EDTA^{4-} , $\text{NCS}^\ominus$, $\text{C}_2\text{O}_4^{2-}$
(3) $\text{NO}_2^\ominus$, $\text{C}_2\text{O}_4^{2-}$, EDTA^{4-} (4) $\text{C}_2\text{O}_4^{2-}$, $\text{NO}_2^\ominus$, $\text{NCS}^\ominus$
CCC316
- One mole of the octahedral complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with excess of AgNO_3 solution to yield two moles of $\text{AgCl}_{(s)}$. The structure of the complex is :
[JEE (Main) 2025]
(1) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ (2) $[\text{Co}(\text{NH}_3)_4\text{Cl}]\cdot\text{Cl}_2\cdot\text{NH}_3$
(3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}\cdot\text{NH}_3$ (4) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]\cdot 2\text{NH}_3$
CCC383
- On treatment of 100 mL of 0.1 M solution of $\text{CoCl}_3 \cdot 6\text{H}_2\text{O}$ with excess AgNO_3 ; 1.2×10^{22} ions are precipitated. The complex is :-
[JEE (Main) 2017]
(1) $[\text{Co}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$ (2) $[\text{Co}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
(3) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$ (4) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
CCC037
- The correct IUPAC name of $[\text{PtBr}_2(\text{PMe}_3)_2]$ is:
[JEE (Main) 2024]
(1) bis(trimethylphosphine)dibromoplatinum(II)
(2) bis[bromo(trimethylphosphine)]platinum(II)
(3) dibromobis(trimethylphosphine)platinum(II)
(4) dibromodi(trimethylphosphine)platinum(II)
CCC345
- The total number of possible isomers for square-planar $[\text{Pt}(\text{Cl})(\text{NO}_2)(\text{NO}_3)(\text{SCN})]^{2-}$ is :-
[JEE (Main) 2018]
(1) 16 (2) 8 (3) 24 (4) 12
CCC040
- Which of the following complex has a possibility to exist as meridional isomer?
[JEE (Main) 2023]
(1) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$ (2) $[\text{Co}(\text{en})_3]$
(3) $[\text{Co}(\text{en})_2\text{Cl}_2]$ (4) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
CCC317
- Consider the following reaction and statements :
 $[\text{Co}(\text{NH}_3)_4\text{Br}_2]^\oplus \text{Br}^\ominus \rightarrow [\text{Co}(\text{NH}_3)_3\text{Br}_3] + \text{NH}_3$
(I) Two isomers are produced if the reactant complex ion is a cis-isomer.
(II) Two isomers are produced if the reactant complex ion is a trans-isomer.
(III) Only one isomer is produced if the reactant complex ion is a trans-isomer.
(IV) Only one isomer is produced if the reactant complex ion is a cis-isomer.
The correct statements are :
[JEE (Main) 2018]
(1) (I) and (III) (2) (III) and (IV) (3) (II) and (IV) (4) (I) and (II)
CCC039

Coordination Compounds

9. The number of geometrical isomers possible in triamminetrinitrocobalt (III) is X and in trioxalatochromate (III) is Y. Then the value of X + Y is _____. [JEE (Main) 2021]

CCC047

10. Which one of the following complexes shows optical isomerism :- [JEE (Main) 2016]

- (1) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (2) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
(3) $\text{cis}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (4) $\text{trans}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
(en = ethylenediamine)

CCC033

11. The d-electronic configuration of an octahedral Co(II) complex having magnetic moment of 3.95 BM is : [JEE (Main) 2025]

- (1) $t_{2g}^6 e_g^1$ (2) $t_{2g}^3 e_g^0$ (3) $t_{2g}^5 e_g^2$ (4) $e^4 t_2^3$

CCC384

12. Number of complexes from the following with even number of unpaired "d" electrons is ____.

$[\text{V}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Ni}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$

[Given atomic numbers : V = 23, Cr = 24, Fe = 26, Ni = 28, Cu = 29]

[JEE (Main) 2024]

- (1) 2 (2) 4 (3) 5 (4) 1

CCC346

13. The 'Spin only' Magnetic moment for $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is ____ $\times 10^{-1}$ BM.
(given = Atomic number of Ni : 28)

[JEE (Main) 2024]

CCC347

14. The ratio of spin-only magnetic moment values $\mu_{\text{eff}}[\text{Cr}(\text{CN})_6]^{3-} / \mu_{\text{eff}}[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ is ____.

[JEE (Main) 2023]

CCC318

15. $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{CoF}_6]^{3-}$ are respectively known as:

[JEE (Main) 2024]

- (1) Spin free Complex, Spin paired Complex
(2) Spin paired Complex, Spin free Complex
(3) Outer orbital Complex, Inner orbital Complex
(4) Inner orbital Complex, Spin paired Complex

CCC348

16. Match List-I with List-II.

List-I (Complex)		List-II (Hybridisation & Magnetic characters)	
(A)	$[\text{MnBr}_4]^{2-}$	(I)	d^2sp^3 & diamagnetic
(B)	$[\text{FeF}_6]^{3-}$	(II)	sp^3d^2 & paramagnetic
(C)	$[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$	(III)	sp^3 & diamagnetic
(D)	$[\text{Ni}(\text{CO})_4]$	(IV)	sp^3 & paramagnetic

Choose the **correct** answer from the options given below :

[JEE (Main) 2025]

- (1) (A)-(III), (B)-(II), (C)-(I), (D)-(IV) (2) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)
(3) (A)-(IV), (B)-(I), (C)-(II), (D)-(III) (4) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

CCC385

17. Identify the coordination complexes in which the central metal ion has d^4 configuration.

- (A) $[\text{FeO}_4]^{2-}$ (B) $[\text{Mn}(\text{CN})_6]^{3-}$ (C) $[\text{Fe}(\text{CN})_6]^{3-}$ (D) $\text{Cr}_2(\text{O}=\text{C}(\text{Me})_2)_2(\text{H}_2\text{O})_2$
 (E) $[\text{NiF}_6]^{2-}$

Choose the **correct** answer from the options given below :

[JEE (Main) 2025]

- (1) (C) and (E) only (2) (B), (C) and (D) only
 (3) (B) and (D) only (4) (A), (B) and (E) only

CCC386

18. Total number of molecules/species from following which will be paramagnetic is ____.

- $\text{O}_2, \text{O}_2^+, \text{O}_2^-, \text{NO}, \text{NO}_2, \text{CO}, \text{K}_2[\text{NiCl}_4],$
 $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3, \text{K}_2[\text{Ni}(\text{CN})_4]$

[JEE (Main) 2025]

CCC387

19. The correct order of spin-only magnetic moments among the following is :

(Atomic number : Mn = 25, Co = 27, Ni = 28, Zn = 30)

[JEE (Main) 2018]

- (1) $[\text{ZnCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
 (2) $[\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
 (3) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{NiCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$
 (4) $[\text{NiCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-} > [\text{ZnCl}_4]^{2-}$

CCC041

20. The pair having the same magnetic moment is :-

[At. No.: Cr = 24, Mn = 25, Fe = 26, Co = 27]

[JEE (Main) 2016]

- (1) $[\text{CoCl}_4]^{2-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (2) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{CoCl}_4]^{2-}$
 (3) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ (4) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$

CCC034

21. The correct increasing order of stability of the complexes based on Δ_0 value is :

- (I) $[\text{Mn}(\text{CN})_6]^{3-}$ (II) $[\text{Co}(\text{CN})_6]^{4-}$ (III) $[\text{Fe}(\text{CN})_6]^{4-}$ (IV) $[\text{Fe}(\text{CN})_6]^{3-}$

[JEE (Main) 2025]

- (1) II < III < I < IV (2) IV < III < II < I (3) I < II < IV < III (4) III < II < IV < I

CCC388

22. The conditions and consequence that favours the t_{2g}^3, e_g^1 configuration in a metal complex are :

[JEE (Main) 2025]

- (1) weak field ligand, high spin complex (2) strong field ligand, high spin complex
 (3) strong field ligand, low spin complex (4) weak field ligand, low spin complex

CCC389

23. Identify the homoleptic complex(es) that is/are low spin.

- (A) $[\text{Fe}(\text{CN})_5\text{NO}]^{2-}$ (B) $[\text{CoF}_6]^{3-}$ (C) $[\text{Fe}(\text{CN})_6]^{4-}$ (D) $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (E) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$

Choose the **correct** answer from the options given below :

[JEE (Main) 2025]

- (1) (B) and (E) only (2) (A) and (C) only (3) (C) and (D) only (4) (C) only

CCC390

24. In which of the following complexes the CFSE, Δ_0 will be equal to zero?

[JEE (Main) 2025]

- (1) $[\text{Fe}(\text{NH}_3)_6]\text{Br}_2$ (2) $[\text{Fe}(\text{en})_3]\text{Cl}_3$ (3) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (4) $\text{K}_3[\text{Fe}(\text{SCN})_6]$

CCC391

25. From the magnetic behaviour of $[\text{NiCl}_4]^{2-}$ (paramagnetic) and $[\text{Ni}(\text{CO})_4]$ (diamagnetic), choose the correct geometry and oxidation state. [JEE (Main) 2025]

- (1) $[\text{NiCl}_4]^{2-}$: Ni^{II} , square planar, $[\text{Ni}(\text{CO})_4]$: $\text{Ni}(0)$, square planar
 (2) $[\text{NiCl}_4]^{2-}$: Ni^{II} , tetrahedral, $[\text{Ni}(\text{CO})_4]$: $\text{Ni}(0)$, tetrahedral
 (3) $[\text{NiCl}_4]^{2-}$: Ni^{II} , tetrahedral, $[\text{Ni}(\text{CO})_4]$: Ni^{II} , square planar
 (4) $[\text{NiCl}_4]^{2-}$: $\text{Ni}(0)$, tetrahedral, $[\text{Ni}(\text{CO})_4]$: $\text{Ni}(0)$, square planar

CCC392

26. Match List-I with List-II.

List-I (Complex)	List-II (Hybridisation of central metal ion)
(A) $[\text{CoF}_6]^{3-}$	(I) d^2sp^3
(B) $[\text{NiCl}_4]^{2-}$	(II) sp^3
(C) $[\text{Co}(\text{NH}_3)_6]^{3+}$	(III) sp^3d^2
(D) $[\text{Ni}(\text{CN})_4]^{2-}$	(IV) dsp^2

Choose the **correct** answer from the options given below :

[JEE (Main) 2025]

- (1) (A)-(I), (B)-(IV), (C)-(III), (D)-(II) (2) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
 (3) (A)-(I), (B)-(II), (C)-(III), (D)-(IV) (4) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

CCC393

27. The correct combination is

[JEE (Main) 2018]

- (1) $[\text{Ni}(\text{CN})_4]^{2-}$ – tetrahedral ; $[\text{Ni}(\text{CO})_4]$ – paramagnetic
 (2) $[\text{NiCl}_4]^{2-}$ – paramagnetic ; $[\text{Ni}(\text{CO})_4]$ – tetrahedral
 (3) $[\text{NiCl}_4]^{2-}$ – diamagnetic ; $[\text{Ni}(\text{CO})_4]$ – square – planar
 (4) $[\text{NiCl}_4]^{2-}$ – square – planar ; $[\text{Ni}(\text{CN})_4]^{2-}$ – paramagnetic

CCC042

28. The correct order of the following complexes in terms of their crystal field stabilization energies is :

[JEE (Main) 2025]

- (1) $[\text{Co}(\text{NH}_3)_4]^{2+} < [\text{Co}(\text{NH}_3)_6]^{2+} < [\text{Co}(\text{en})_3]^{3+} < [\text{Co}(\text{NH}_3)_6]^{3+}$
 (2) $[\text{Co}(\text{NH}_3)_4]^{2+} < [\text{Co}(\text{NH}_3)_6]^{2+} < [\text{Co}(\text{NH}_3)_6]^{3+} < [\text{Co}(\text{en})_3]^{3+}$
 (3) $[\text{Co}(\text{NH}_3)_6]^{2+} < [\text{Co}(\text{NH}_3)_6]^{3+} < [\text{Co}(\text{NH}_3)_4]^{2+} < [\text{Co}(\text{en})_3]^{3+}$
 (4) $[\text{Co}(\text{en})_3]^{3+} < [\text{Co}(\text{NH}_3)_6]^{3+} < [\text{Co}(\text{NH}_3)_6]^{2+} < [\text{Co}(\text{NH}_3)_4]^{2+}$

CCC394

29. Consider the following complexes.

$[\text{CoCl}(\text{NH}_3)_5]^{2+}$, (A)	$[\text{Co}(\text{CN})_6]^{3-}$, (B)
$[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$, (C)	$[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ (D)

The correct order of A, B, C and D in terms of wavenumber of light absorbed is :

[JEE (Main) 2024]

- (1) $C < D < A < B$ (2) $D < A < C < B$
 (3) $A < C < B < D$ (4) $B < C < A < D$

CCC349

30. Arrange the following Cobalt complexes in the order of increasing Crystal Field Stabilization Energy (CFSE) value. [JEE (Main) 2021]

Complexes : $[\text{CoF}_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{Co}(\text{en})_3]^{3+}$
 A B C D

Choose the correct option :

- (1) $A < B < C < D$ (2) $B < A < C < D$ (3) $B < C < D < A$ (4) $C < D < B < A$

CCC048

31. Which one of the following metal complexes is most stable? [JEE (Main) 2021]
 (1) $[\text{Co(en)}(\text{NH}_3)_4]\text{Cl}_2$ (2) $[\text{Co(en)}_3]\text{Cl}_2$
 (3) $[\text{Co(en)}_2(\text{NH}_3)_2]\text{Cl}_2$ (4) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_2$
 CCC046
32. Identify the **CORRECT** trend given below : [JEE (Main) 2016]
 (Atomic No. = Ti : 22, Cr : 24 and Mo : 42)
 (1) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} < [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} < [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
 (2) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} < [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} > [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
 (3) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} > [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} > [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
 (4) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} > [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} < [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
 CCC035
33. When Ethane-1,2-diamine is added progressively to an aqueous solution of Nickel (II) chloride, the sequence of colour change observed will be : [JEE (Main) 2025]
 (1) Pale Blue $\rightarrow$ Blue $\rightarrow$ Green $\rightarrow$ Violet (2) Pale Blue $\rightarrow$ Blue $\rightarrow$ Violet $\rightarrow$ Green
 (3) Green $\rightarrow$ Pale Blue $\rightarrow$ Blue $\rightarrow$ Violet (4) Violet $\rightarrow$ Blue $\rightarrow$ Pale Blue $\rightarrow$ Green
 CCC395
34. Among the statements(a)-(d), the incorrect ones are-
 (a) Octahedral Co(III) complexes with strong field ligands have very high magnetic moments
 (b) When $\Delta_0 < P$, the d-electron configuration of Co(III) in an octahedral complex is $t_{2g}^4 e_g^2$
 (c) Wavelength of light absorbed by $[\text{Co(en)}_3]^{3+}$ is lower than that of $[\text{CoF}_6]^{3-}$
 (d) If the Δ_0 for an octahedral complex of Co(III) is $18,000 \text{ cm}^{-1}$, the Δ_t for its tetrahedral complex with the same ligand will be $16,000 \text{ cm}^{-1}$ [JEE (Main) 2020]
 (1) (a) and (b) only (2) (c) and (d) only (3) (b) and (c) only (4) (a) and (d) only
 CCC045
35. Two complexes, $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (A) and $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ (B) are violet and yellow coloured, respectively. The incorrect statement regarding them is : [JEE (Main) 2019]
 (1) Δ_0 value of (A) is less than that of (B).
 (2) Δ_0 value of (A) and (B) are calculated from the energies of violet and yellow light, respectively
 (3) Both absorb energies corresponding to their complementary colors.
 (4) Both are paramagnetic with three unpaired electrons.
 CCC044
36. $[\text{Co}_2(\text{CO})_8]$ displays :- [JEE (Main) 2017]
 (1) no Co-Co bond, four terminal CO and four bridging CO
 (2) one Co-Co bond, six terminal CO and two bridging CO
 (3) no Co-Co bond, six terminal CO and two bridging CO
 (4) one Co-Co bond, four terminal CO and four bridging CO
 CCC036
37. Low oxidation state of metals in their complexes are common when ligands: [JEE (Main) 2022]
 (1) have good π -accepting character (2) have good σ -donor character
 (3) are having good π -donating ability (4) are having poor σ -donating ability
 CCC049

EXERCISE - JEE (Advanced) PYQ

- EDTA⁴⁻ is ethylenediaminetetraacetate ion. The total number of N-Co-O bond angles in [Co(EDTA)]¹⁻ complex ion is [JEE (Advanced) 2013]
CCC073
- Among V(CO)₆, Cr(CO)₅, Cu(CO)₃, Mn(CO)₅, Fe(CO)₅, [Co(CO)₃]³⁻, [Cr(CO)₄]⁴⁻, and Ir(CO)₃, the total number of species isoelectronic with Ni(CO)₄ is _____.
[Given, atomic number : V = 23, Cr = 24, Mn = 25, Fe = 26, Co = 27, Ni = 28, Cu = 29, Ir = 77] [JEE (Advanced) 2024]
CCC350
- The complex(es), which can exhibit the type of isomerism shown by [Pt(NH₃)₂Br₂], is(are)
[en = H₂NCH₂CH₂NH₂] [JEE (Advanced) 2023]
(A) [Pt(en)(SCN)₂] (B) [Zn(NH₃)₂Cl₂]
(C) [Pt(NH₃)₂Cl₄] (D) [Cr(en)₂(H₂O)(SO₄)]⁺ CCC320
- The total number of possible isomers for [Pt(NH₃)₄Cl₂]Br₂ is ____ [JEE (Advanced) 2021]
CCC090
- The pair(s) of coordination complex/ion exhibiting the same kind of isomerism is(are) - [JEE (Advanced) 2013]
(A) [Cr(NH₃)₅Cl]Cl₂ and [Cr(NH₃)₄Cl₂]Cl
(B) [Co(NH₃)₄Cl₂]⁺ and [Pt(NH₃)₂(H₂O)Cl]⁺
(C) [CoBr₂Cl₂]²⁻ and [PtBr₂Cl₂]²⁻
(D) [Pt(NH₃)₃(NO₃)] Cl and [Pt(NH₃)₃Cl] Br CCC072
- Among the following options, select the option in which each complex in Set-I shows geometrical isomerism and the two complexes in Set-II are ionization isomers of each other.
[en = H₂NCH₂CH₂NH₂] [JEE (Advanced) 2024]
(A) Set-I : [Ni(CO)₄] and [PdCl₂(PPh₃)₂]
Set-II : [Co(NH₃)₅Cl] SO₄ and [Co(NH₃)₅(SO₄)]Cl
(B) Set-I : [Co(en)(NH₃)₂Cl₂] and [PdCl₂(PPh₃)₂]
Set-II : [Co(NH₃)₆] [Cr(CN)₆] and [Cr(NH₃)₆] [Co(CN)₆]
(C) Set-I : [Co(NH₃)₃(NO₂)₃] and [Co(en)₂Cl₂]
Set-II : [Co(NH₃)₅Cl]SO₄ and [Co(NH₃)₅(SO₄)]Cl
(D) Set-I : [Cr(NH₃)₅Cl]Cl₂ and [Co(en)(NH₃)₂Cl₂]
Set-II : [Cr(H₂O)₆]Cl₃ and [Cr(H₂O)₅Cl]Cl₂ · H₂O CCC351
- Total number of cis N-Mn-Cl bond angles (that is, Mn-N and Mn-Cl bonds in cis positions) present in a molecule of cis-[Mn(en)₂Cl₂] complex is ____ (en = NH₂CH₂CH₂NH₂) [JEE (Advanced) 2019]
CCC086
- The number of geometric isomers possible for the complex [CoL₂Cl₂]⁺ (L = H₂NCH₂CH₂O⁺) is [JEE (Advanced) 2016]
CCC079
- Among the complex ions,
[Co(NH₂-CH₂-CH₂-NH₂)₂Cl₂]⁺, [CrCl₂(C₂O₄)₂]³⁻, [Fe(H₂O)₄(OH)₂]⁺, [Fe(NH₃)₂(CN)₄]⁻,
[Co(NH₂-CH₂-CH₂-NH₂)₂(NH₃)Cl]²⁺ and [Co(NH₃)₄(H₂O)Cl]²⁺, the number of complex ion(s) that show(s) cis-trans isomerism is - [JEE (Advanced) 2015]
CCC077

10. Match each coordination compound in List-I with an appropriate pair of characteristics from List-II and select the correct answer using the code given below the lists. [JEE (Advanced) 2014]
 {en = $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$; atomic numbers ; Ti = 22 ; Cr = 24 ; Co = 27 ; Pt = 78}

List-I

- (P) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
 (Q) $[\text{Ti}(\text{H}_2\text{O})_5\text{Cl}](\text{NO}_3)_2$
 (R) $[\text{Pt}(\text{en})(\text{NH}_3)\text{Cl}]\text{NO}_3$
 (S) $[\text{Co}(\text{NH}_3)_4(\text{NO}_3)_2]\text{NO}_3$

Code :

- | | P | Q | R | S |
|-----|---|---|---|---|
| (A) | 4 | 2 | 3 | 1 |
| (B) | 3 | 1 | 4 | 2 |
| (C) | 2 | 1 | 3 | 4 |
| (D) | 1 | 3 | 4 | 2 |

List-II

- (1) Paramagnetic and exhibits ionisation isomerism
 (2) Diamagnetic and exhibits cis-trans isomerism
 (3) Paramagnetic and exhibits cis-trans isomerism
 (4) Diamagnetic and exhibits ionisation isomerism

CCC074

11. Among the following complexes, the total number of diamagnetic species is ____.



[Given, atomic number : Mn = 25, Fe = 26, Co = 27;

en = $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$]

[JEE (Advanced) 2024]

CCC352

12. LIST-I contains metal species and LIST-II contains their properties.

LIST-I

- (I) $[\text{Cr}(\text{CN})_6]^{4-}$
 (II) $[\text{RuCl}_6]^{2-}$
 (III) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$
 (IV) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$

LIST-II

- (P) t_{2g} orbitals contain 4 electrons
 (Q) $\mu(\text{spin-only}) = 4.9 \text{ BM}$
 (R) low spin complex ion
 (S) metal ion in +4 oxidation state
 (T) d^4 species

[Given : Atomic number of Cr = 24, Ru = 44, Fe = 26]

[JEE (Advanced) 2022]

Match each metal species in LIST-I with their properties in LIST-II, and choose the correct option

- (A) I $\rightarrow$ R, T; II $\rightarrow$ P, S; III $\rightarrow$ Q, T; IV $\rightarrow$ P, Q
 (B) I $\rightarrow$ R, S; II $\rightarrow$ P, T; III $\rightarrow$ P, Q; IV $\rightarrow$ Q, T
 (C) I $\rightarrow$ P, R; II $\rightarrow$ R, S; III $\rightarrow$ R, T; IV $\rightarrow$ P, T
 (D) I $\rightarrow$ Q, T; II $\rightarrow$ S, T; III $\rightarrow$ P, T; IV $\rightarrow$ Q, R

CCC092

13. The calculated spin only magnetic moments of $[\text{Cr}(\text{NH}_3)_6]^{3+}$ and $[\text{CuF}_6]^{3-}$ in BM, respectively, are (Atomic numbers of Cr and Cu are 24 and 29, respectively)

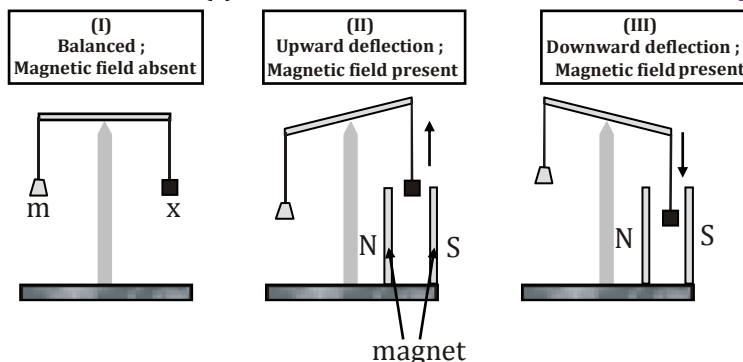
[JEE (Advanced) 2021]

- (A) 3.87 and 2.84 (B) 4.90 and 1.73 (C) 3.87 and 1.73 (D) 4.90 and 2.84

CCC091

14. In an experiment, m grams of a compound X (gas/liquid/solid) taken in a container is loaded in a balance as shown in figure I below. In the presence of a magnetic field, the pan with X is either deflected upwards (figure II), or deflected downwards (figure III), depending on the compound X. Identify the correct statement (s)

[JEE (Advanced) 2020]



- (A) If X is $\text{H}_2\text{O}(\text{l})$, deflection of the pan is upwards.
 (B) If X is $\text{K}_4[\text{Fe}(\text{CN})_6](\text{s})$, deflection of the pan is upwards.
 (C) If X is $\text{O}_2(\text{g})$, deflection of the pan is downwards.
 (D) If X is $\text{C}_6\text{H}_6(\text{l})$, deflection of the pan is downwards.

CCC088

15. Addition of excess aqueous ammonia to a pink coloured aqueous solution of $\text{MCl}_2 \cdot 6\text{H}_2\text{O}$ (X) and NH_4Cl gives an octahedral complex Y in the presence of air. In aqueous solution, complex Y behaves as 1 : 3 electrolyte. The reaction of X with excess HCl at room temperature results in the formation of a blue coloured complex Z. The calculated spin only magnetic moment of X and Z is 3.87 B.M., whereas it is zero for complex Y.

Among the following options, which statements is(are) **CORRECT** ?

[JEE (Advanced) 2017]

- (A) The hybridization of the central metal ion in Y is d^2sp^3
 (B) Z is tetrahedral complex
 (C) Addition of silver nitrate to Y gives only two equivalents of silver chloride
 (D) When X and Z are in equilibrium at 0°C , the colour of the solution is pink

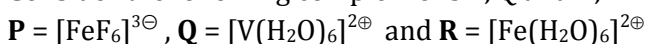
CCC081

16. For the octahedral complexes of Fe^{3+} in SCN^- (thiocyanato – S) and in CN^- ligand environments, the difference between the spin only magnetic moments in Bohr magnetons (when approximated to the nearest integer) is : [Atomic number of Fe = 26]

[JEE (Advanced) 2015]

CCC075

17. Consider the following complex ions P, Q and R,



The correct order of the complex ions, according to their spin-only magnetic moment values (in B.M.) is -

[JEE (Advanced) 2013]

- (A) $\text{R} < \text{Q} < \text{P}$ (B) $\text{Q} < \text{R} < \text{P}$ (C) $\text{R} < \text{P} < \text{Q}$ (D) $\text{Q} < \text{P} < \text{R}$

CCC071

18. Match the electronic configurations in List-I with appropriate metal complex ions in List-II and choose the correct option.

[Atomic Number: Fe = 26, Mn = 25, Co = 27]

[JEE (Advanced) 2023]

List-I

- (P) $t_{2g}^6 e_g^0$
 (Q) $t_{2g}^3 e_g^2$
 (R) $e^2 t_{2g}^3 s$
 (S) $t_{2g}^4 e_g^2$

List-II

- (1) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 (2) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$
 (3) $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (4) $[\text{FeCl}_4]^-$
 (5) $[\text{CoCl}_4]^{2-}$

- (A) $\text{P} \rightarrow 1; \text{Q} \rightarrow 4; \text{R} \rightarrow 2; \text{S} \rightarrow 3$
 (C) $\text{P} \rightarrow 3; \text{Q} \rightarrow 2; \text{R} \rightarrow 5; \text{S} \rightarrow 1$

- (B) $\text{P} \rightarrow 1; \text{Q} \rightarrow 2; \text{R} \rightarrow 4; \text{S} \rightarrow 5$
 (D) $\text{P} \rightarrow 3; \text{Q} \rightarrow 2; \text{R} \rightarrow 4; \text{S} \rightarrow 1$

CCC319

19. The pair(s) of complexes wherein both exhibit tetrahedral geometry is(are)

(Note: py = pyridine)

[JEE (Advanced) 2021]

Given: Atomic numbers of Fe, Co, Ni and Cu are 26, 27, 28 and 29, respectively)

- (A) $[\text{FeCl}_4]^-$ and $[\text{Fe}(\text{CO})_4]^{2-}$ (B) $[\text{Co}(\text{CO})_4]^\ominus$ and $[\text{CoCl}_4]^{2-}$
 (C) $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ (D) $[\text{Cu}(\text{py})_4]^\oplus$ and $[\text{Cu}(\text{CN})_4]^{3-}$

CCC089

20. Choose the correct statement(s) among the following :

[JEE (Advanced) 2020]

- (A) $[\text{FeCl}_4]^\ominus$ has tetrahedral geometry.
 (B) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^\oplus$ has 2 geometrical isomers.
 (C) $[\text{FeCl}_4]^\ominus$ has higher spin – only magnetic moment than $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^\oplus$.
 (D) The cobalt ion in $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]$ has sp^3d^2 hybridization.

CCC087

21. Among the species given below, the total number of diamagnetic species is ____.
H atom, NO₂ monomer, O₂[⊖] (superoxide), dimeric sulphur in vapour phase,
Mn₃O₄, (NH₄)₂[FeCl₄], (NH₄)₂[NiCl₄], K₂MnO₄, K₂CrO₄ [JEE (Advanced) 2018] CCC085
22. Among [I₃]⁺, [SiO₄]⁴⁻, SO₂Cl₂, XeF₂, SF₄, ClF₃, Ni(CO)₄, XeO₂F₂, [PtCl₄]²⁻, XeF₄, and SOCl₂, the total number of species having sp³ hybridised central atom is _____. [JEE (Advanced) 2023] CCC353
23. Match each set of hybrid orbitals from LIST-I with complex (es) given in LIST-II. [JEE (Advanced) 2018]
- | LIST-I | LIST-II |
|------------------------------------|--|
| (P) dsp ² | (1) [FeF ₆] ⁴⁻ |
| (Q) sp ³ | (2) [Ti(H ₂ O) ₃ Cl ₃] |
| (R) sp ³ d ² | (3) [Cr(NH ₃) ₆] ³⁺ |
| (S) d ² sp ³ | (4) [FeCl ₄] ²⁻ |
| | (5) Ni(CO) ₄ |
| | (6) [Ni(CN) ₄] ²⁻ |
- The correct option is :-
- | | |
|------------------------------------|--------------------------------------|
| (A) P → 5; Q → 4,6; R → 2,3; S → 1 | (B) P → 5,6; Q → 4; R → 3; S → 1,2 |
| (C) P → 6; Q → 4,5; R → 1; S → 2,3 | (D) P → 4,6; Q → 5,6; R → 1,2; S → 3 |
- CCC084
24. The geometries of the ammonia complexes of Ni²⁺, Pt²⁺ and Zn²⁺, respectively, are : [JEE (Advanced) 2016]
- (A) octahedral, square planar and tetrahedral
(B) square planar, octahedral and tetrahedral
(C) tetrahedral, square planar and octahedral
(D) octahedral, tetrahedral and square planar
- CCC080
25. Among [Ni(CO)₄], [NiCl₄]²⁻, [Co(NH₃)₄Cl₂]Cl, Na₃[CoF₆], Na₂O₂ and CsO₂, the total number of paramagnetic compounds is - [JEE (Advanced) 2015]
- (A) 2 (B) 3 (C) 4 (D) 5
- CCC078
26. The correct option(s) regarding the complex [Co(en)(NH₃)₃(H₂O)]³⁺ :-
(en = H₂NCH₂CH₂NH₂) is (are) [JEE (Advanced) 2018]
- (A) It has two geometrical isomers
(B) It will have three geometrical isomers if bidentate 'en' is replaced by two cyanide ligands
(C) It is paramagnetic
(D) It absorbs light at longer wavelength as compared to [Co(en)(NH₃)₄]³⁺
- CCC083
27. In the complex acetyl bromidodicarbonylbis (triethyl phosphine) iron(II), the number of Fe-C bond(s) is- [JEE (Advanced) 2015] CCC076
28. The correct statement(s) regarding the binary transition metal carbonyl compounds is (are)
(Atomic numbers : Fe = 26, Ni = 28) [JEE (Advanced) 2018]
- (A) Total number of valence shell electrons at metal centre in Fe(CO)₅ or Ni(CO)₄ is 16
(B) These are predominantly low spin in nature
(C) Metal - carbon bond strengthens when the oxidation state of the metal is lowered
(D) The carbonyl C-O bond weakens when the oxidation state of the metal is increased
- CCC082

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	C	B	C	B	C	D	C	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	D	A	C	B	A	B	D	D	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	C	A	D	D	B	A	C	C	B
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	A	C	B	A	B	B	D	B	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	D	D	A	B	C	A	B	D	D
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	B	A	A	B	A	D	B	D	B	B
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	D	B	B	C	D	B	A	D	D	B
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	C	D	C	A	C	C	A	C	C	B
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	B	C	D	D	D	B	C	C	B	D
Que.	91	92	93	94	95	96	97	98	99	100
Ans.	C	D	A	B	A	C	A	C	D	A
Que.	101	102	103	104	105	106	107	108	109	110
Ans.	C	C	D	B	C	C	D	C	B	B
Que.	111	112	113	114	115	116	117	118	119	120
Ans.	A	D	A	D	A	B	B	A	C	D
Que.	121	122	123	124	125	126	127	128	129	130
Ans.	A	A	C	B	C	D	B	B	C	A
Que.	131	132	133	134	135	136	137	138	139	140
Ans.	B	A	D	B	C	C	D	B	A	C
Que.	141	142	143	144	145	146				
Ans.	C	B	D	D	C	B				

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ABD	AB	AC	BC	ABCD	AC	CD	BCD	AB	AB
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	ABCD	AB	BD	AC	ACD	ABD	BD	ABC	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	C	B	C	D	B	C	A	C
Que.	31	32	33	34	35	36				
Ans.	D	C	A	A	B	A				

EXERCISE (S)

Que.	1	2	3	4	5	6	7	8		
Ans.	3	34	5	4	2	33	36	(a) 4, (b) 9, (c) 4, (d) 4, (e) 5		
Que.	9	10	11	12	13	14	15	16	17	18
Ans.	3	0	2	15	2	9	2	2	6	5
Que.	19	20	21	22	23	24	25	26	27	28
Ans.	2	5	2	26.92	0.30	3.50	8.00	1.33	2.83	2.25
Que.	29	30								
Ans.	0.00	0.33								

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	14	1	1	1	3	4	1	1	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	28	1	2	4	3	6	3	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	1	3	4	2	2	2	2	2	2
Que.	31	32	33	34	35	36	37			
Ans.	2	2	3	4	2	2	1			

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	8	3	CD	6	BD	C	6	5	6	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	A	A	ABC	ABD	4	B	D	ABD	AC
Que.	21	22	23	24	25	26	27	28		
Ans.	1	5	C	A	B	ABD	3	BC		